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Master Thesis

Easy-to-Read (E2R) Adaptation of Figurative Language

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Abstract

Figurative language, such as idioms and metaphors, is a barrier to text accessibility. Easy-to-Read guidelines, which serve readers with intellectual disabilities, low literacy, or limited language proficiency, instruct writers to avoid figurative language but give no method for rewriting it. This thesis studies whether large language models can fill that gap by detecting figurative expressions and rewriting them into literal, meaning-preserving, easier-to-read English, without task-specific training.

We build an agentic pipeline that decomposes the task into three explicit, inspectable steps (detection, explanation, and transformation) and compare it against a monolithic single-prompt baseline. Both run on open-weights models at two scales, an 8B deliverable and a 70B comparator, and are evaluated on SemEval-2022 Task 2 idioms and the VU Amsterdam Metaphor Corpus metaphors, using automatic detection metrics and a Direct Assessment human-evaluation survey for replacement quality.

The central finding is that reliable quality comes from explicit task structure rather than from model scale or prompt wording. Scale lifts detection but not human-perceived replacement quality: in this setup, scale did not buy, and on simplicity may have cost, human-perceived quality. Gains from prompt wording do not transfer across model sizes and can reverse in sign. Decomposition yields a small but consistent advantage on meaning preservation, falling just short of statistical significance, with a tied composite score; its clearer benefit is that each failure can be traced to a named step. Across all systems the replacements preserve meaning but do not simplify, which marks the central remaining challenge for Easy-to-Read adaptation, and shows that explicit structure, together with human-grounded evaluation, is the lever that can be relied on in this task.

The thesis contributes an agentic open-weights pipeline for this task with a monolithic baseline as a designed control, an adapted Direct Assessment protocol for evaluating figurative-to-literal replacement, and a by-product dataset of 108 human-written alternative replacements with rubric ratings for 81 system outputs.

Resumen

El lenguaje figurado, como los modismos y las metáforas, constituye una barrera para la accesibilidad de los textos. Las pautas de Lectura Fácil, dirigidas a personas con discapacidad intelectual, baja alfabetización o competencia lingüística limitada, indican que debe evitarse el lenguaje figurado, pero no ofrecen un método para reescribirlo. Esta tesis estudia si los grandes modelos de lenguaje pueden cubrir esa carencia, detectando expresiones figuradas y reescribiéndolas en un inglés literal, que preserve el significado y resulte más fácil de leer, sin entrenamiento específico para la tarea.

Se desarrolla una arquitectura agéntica que descompone la tarea en tres pasos explícitos e inspeccionables (detección, explicación y transformación) y se compara con una línea base monolítica de un solo prompt. Ambas se ejecutan sobre modelos de pesos abiertos en dos escalas, un modelo de 8B como sistema final y uno de 70B como comparador, y se evalúan sobre los modismos de la tarea SemEval-2022 Task 2 y las metáforas del VU Amsterdam Metaphor Corpus, mediante métricas automáticas de detección y una encuesta de evaluación humana de tipo Direct Assessment para la calidad de la sustitución.

El hallazgo central es que la calidad fiable proviene de la estructura explícita de la tarea, y no de la escala del modelo ni de la formulación del prompt. La escala mejora la detección, pero no la calidad de la sustitución percibida por las personas: en esta configuración, la escala no aportó, y en simplicidad pudo incluso restar, calidad percibida por las personas. Las mejoras derivadas de la formulación del prompt no se transfieren entre escalas de modelo y pueden invertir su signo. La descomposición ofrece una ventaja direccional y consistente en la preservación del significado que se aproxima al umbral convencional de significación sin alcanzarlo, con una puntuación compuesta empatada, mientras que su beneficio más claro es que cada fallo puede atribuirse a un paso concreto. En todos los sistemas las sustituciones preservan el significado pero no simplifican, lo que señala el principal reto pendiente para la adaptación a Lectura Fácil y muestra que la estructura explícita, junto con una evaluación anclada en el juicio humano, es la palanca fiable en esta tarea.

La tesis aporta una arquitectura agéntica de pesos abiertos para esta tarea, con una línea base monolítica como control de diseño; un protocolo Direct Assessment adaptado para evaluar la sustitución de lo figurado a lo literal; y, como subproducto, un conjunto de datos de 108 sustituciones alternativas redactadas por personas, junto con valoraciones por rúbrica de 81 salidas del sistema.

Acknowledgement

I would like to thank my supervisor, Mari Carmen Suárez-Figueroa, at the Universidad Politécnica de Madrid, for her guidance, her careful feedback, and her patience throughout this work. Her steady direction shaped both the questions this thesis asks and the way it answers them.

I am grateful to 2Hero AB, where this work took shape alongside my internship, for the software infrastructure on which the system was built and for the room to pursue this research in parallel with my work there.

I also thank the volunteers who took part in the human-evaluation survey. The central results of this thesis rest on the time and care they gave, and I am indebted to each of them.

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Declaration on the Use of AI Tools

In preparing this thesis I used AI-based assistants, principally Anthropic's Claude (through the Claude Code interface), as a writing and engineering aid. Their use was limited to drafting and editing prose under my direction, generating figures and helper scripts, checking citations and internal consistency against my own sources, and assisting with the orchestration of experiments.

The research itself is my own: the system design and implementation, the experimental setup and its execution, the analysis and interpretation of the results, and all claims and conclusions. I reviewed every AI-assisted output, verified it against the primary sources and the data, and I take full responsibility for the content of this thesis, including any errors.

Large language models are also an object of study in this work. That experimental use is described in the relevant chapters and is separate from the writing-assistance use declared here.

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List of Acronyms

Acronym	Expansion
AWQ	Activation-aware Weight Quantization
BLEU	Bilingual Evaluation Understudy (n-gram overlap metric)
CoT	Chain-of-Thought
DA	Direct Assessment (continuous human-evaluation protocol)
dtr	detect-then-replace (single-call task variant)
E2R	Easy-to-Read
F1	F_1 score (harmonic mean of precision and recall)
IoU	Intersection over Union (span-overlap criterion)
LLM	Large Language Model
MIP	Metaphor Identification Procedure
MIPVU	Metaphor Identification Procedure, VU University Amsterdam
MWE	Multi-Word Expression
QC	Quality Control
RQ	Research Question
SemEval	Semantic Evaluation (shared-task workshop series)
VUAMC	VU Amsterdam Metaphor Corpus

1 Introduction

1.1 Motivation

Easy-to-Read English is a way of writing meant for people who find standard text hard to follow, including people with intellectual disabilities, people with low literacy, and people reading in a language that is not their first. The guidelines for this style share one rule: avoid figurative language. Idioms and metaphors are meant to be removed, because their meaning is not carried by the words themselves, so a reader who takes them literally is misled and a reader who cannot recover the intended sense is locked out altogether. In this setting, figurative language is an accessibility barrier.

The guidelines say to remove such language, but they do not say how to do it, and they leave the rewriting to a human editor. Recent work at UPM has begun to automate Easy-to-Read adaptation one guideline at a time, with a dedicated module for morphological features [28] and another for dialogue structure [7], part of a research line on guideline-driven adaptation whose support application, FACILE, is described in [27]. Figurative language is one of the guidelines this programme has not yet covered, and this thesis is its module: a system that detects an idiom or metaphor and rewrites it into plain, literal English.

The task sits between two research communities that do not meet. Work on figurative language in natural-language processing has built strong systems for finding and labelling idioms and metaphors, but it stops at detection and does not rewrite the text for a reader. Work on automatic text simplification and Easy-to-Read adaptation rewrites text for accessibility, but it handles other guidelines and treats figurative language as out of scope. Neither side addresses the point where they meet, detecting a figurative expression and replacing it with an accessible literal form, and Chapter 2 describes this gap in more detail.

Large language models are what make the rewriting half of the task possible now. Detection alone can be learned from labelled data, and supervised systems for it already exist, but replacement cannot be learned the same way: there is no training set of figurative sentences paired with accessible literal rewrites, and, as this thesis shows, no automatic metric that can score such a rewrite reliably. A model that can produce a literal restatement from zero or one example is therefore the only practical approach to the generative half of the task, and this is the capability that current large language models supply.

1.2 Problem Statement

The task this thesis studies is the following. Given an English sentence, detect any idiomatic or metaphorical expression in it, and produce a restatement that keeps the original meaning, removes the figurative wording, and is easier to read. Detection and rewriting are treated as one connected job, not two separate ones, because a rewrite is only useful if it lands on the right expression and a detection is only useful if something then rewrites it.

Idioms and metaphors are handled in parallel, not as one phenomenon. They share an architecture but never a prompt: the system runs separate workflows for each, because an idiom has a fixed conventional meaning while a metaphor is built from an open mapping between two ideas, and the two call for different handling. The system is built once and applied twice, with the difference between the two phenomena kept explicit rather than blended away.

The central claim of this thesis concerns where reliable quality in this task comes from. Common practice expects two things to drive it: a larger model and a better-worded prompt. The evidence here points elsewhere. Reliable quality comes from explicit structure, that is, from breaking the task into inspectable steps and from grounding the evaluation in human judgement. Model size and prompt wording are levers whose effect cannot be trusted to hold: their gains vary across models and can reverse in sign, and the automatic metrics that might catch the reversal cannot see replacement quality at all. Explicit task structure is the lever whose effect can be observed, attributed to a named step, and trusted. Human evaluation is not itself a lever on quality; it is how that quality is made visible. This is what the thesis means by structure over scale: not that structure always wins, but that it is the lever that can be relied on to mean what it says (Section 8.1).

1.3 Research Questions and Hypotheses

The thesis is organised around three research questions.

RQ1 (Detection and interpretation). To what extent can a zero-/one-shot agentic LLM pipeline accurately detect and interpret idiomatic and metaphorical expressions in context, compared to established supervised benchmarks?

RQ2 (Agentic decomposition). Does an explicit agentic decomposition (detection, then explanation, then transformation) improve figurative-language handling compared to monolithic prompt-based LLM approaches?

RQ3 (Replacement quality). How effectively can an agentic LLM system replace idiomatic and metaphorical expressions with literal, meaning-preserving alternatives, and what trade-offs arise between semantic fidelity and readability?

Four hypotheses make these questions testable.

H1. A zero-/one-shot agentic LLM pipeline can reach performance comparable to supervised systems on idiom detection and metaphor identification, despite using no task-specific training.

H2. Agentic decomposition improves figurative-language replacement quality compared to single-prompt baselines, particularly for metaphors.

H3. Idiomatic expressions are replaced with meaning-preserving literal alternatives more reliably than conceptual metaphors, because metaphors rest on an open mapping between an abstract idea and a concrete one.

H4. Explicit intermediate explanations generated by the agent correlate positively with the semantic correctness of the final literal replacement.

H1 through H3 are tested in the empirical chapters. H4 is not tested in this thesis. Measuring it requires a separate study that scores explanation quality and replacement correctness independently and then correlates them, and that study was out of scope here. H4 is kept on the record because it names the next natural step, and the design needed to test it is set out as future work in Section 8.6.

A fourth concern, whether the system's steps are observable enough to explain its own outputs, runs through the work but is not posed as a research question. It is a property of the design rather than a hypothesis to confirm or reject, so it is examined qualitatively in the Discussion (Section 8.4) rather than tested.

1.4 Contributions

This thesis makes four contributions.

1. An agentic open-weights pipeline for figurative-language Easy-to-Read adaptation, built on FastAPI, LangGraph, and vLLM, with all experiments persisted and reproducible, and a monolithic single-prompt system as a designed control.

2. An adapted Direct Assessment human-evaluation method for figurative-language replacement (Chapter 4), building on the established Direct Assessment protocol [12, 10, 13] with its departures from that protocol documented, since no automatic metric scores this task reliably.

3. Empirical findings on what does and does not transfer in this task: scale lifts detection but not human-perceived replacement quality, prompt gains can reverse in sign across model sizes, and decomposition buys a consistent directional advantage on meaning preservation.

4. A by-product dataset of 108 human-written alternative replacements together with rubric ratings for 81 system outputs, available for future work on this task.

1.5 Document Structure

The rest of the thesis is organised as follows. Chapter 2 reviews the two literatures the task sits between, figurative-language processing and Easy-to-Read adaptation, and closes by stating the gap between them. Chapter 3 describes the agentic pipeline and the monolithic baseline it is compared against. Chapter 4 sets out the evaluation method, including the adapted Direct Assessment survey and the automatic metrics, and explains why the human survey is required rather than optional. The three empirical chapters answer one research question each: Section 5 reports detection and the effect of scale (RQ1), Section 6 reports the agentic-versus-monolithic comparison (RQ2), and Section 7 reports replacement quality and the survey results (RQ3). Chapter 8 assembles these results into the central claim and states the limits of the evidence, and Chapter 9 closes with the answers to the research questions and the directions they open.

The three empirical chapters can be read as a test of three levers on quality. RQ1 tests scale, by comparing an 8B model with a 70B one. RQ2 tests structure, by comparing a decomposed pipeline with a single prompt. RQ3 tests the task itself, by measuring how well the rewriting holds meaning while making the text easier to read, and where those two goals pull against each other. Read together, they show that the lever that pays is structure.

2 Background & Related Work

2.1 Figurative Language

Figurative language covers expressions whose intended meaning is not the literal sum of their words. This thesis works with two kinds. An idiom is a fixed multi-word expression whose meaning is conventional and cannot be read off its parts: "spill the beans" means reveal a secret, and nothing about beans explains why. A metaphor describes one thing in terms of another, mapping a familiar idea onto a less familiar one, as in "a flood of complaints". The two differ in how fixed they are: idioms are largely settled units that a speaker either knows or does not, while metaphors range from dead and conventional ("the leg of a table") to fresh and novel, invented on the spot. This spread, from fully conventional to fully novel, matters for the task, because a conventional expression has a stable meaning to recover while a novel one does not.

Metaphor is annotated at the level of the individual word. The Metaphor Identification Procedure, and its extension MIPVU, marks a word as metaphorical when its meaning in context contrasts with a more basic, concrete meaning that the contextual sense can be derived from [26]. The VU Amsterdam Metaphor Corpus applies this procedure at scale and is the standard resource for metaphor detection (Section 2.5). Idioms, by contrast, are annotated as whole multi-word units that are either idiomatic or literal in a given sentence.

Detection has progressed, but its reported scores can mislead. An analysis of metaphor-recognition models finds that much of their apparent success is conventionalised word-sense disambiguation rather than metaphor competence: systems do well on conventional metaphors, which a model can resolve by memorised word senses, and substantially worse on unconventional ones [22]. A high F-score therefore does not show that a system understands metaphor, a point that frames the supervised comparison in Chapter 5.

Whether large language models genuinely handle figurative language is an open debate, and the evidence pulls in both directions. On the optimistic side, GPT-4 has shown an emergent ability to interpret novel literary metaphors, with its readings rated above those of college students by blind judges [16]. On the sceptical side, models remain weak at detection: one study found that the best system recovered only six of twenty metaphorical words, arguing that detection needs phrase-level analysis and world knowledge that the training data does not supply [8]. Benchmark results sit in between: on the Fig-QA dataset, language models score well above chance but below human level, and the gap is largest in the zero- and few-shot settings this thesis uses [20]. The pattern that reconciles these results is that interpreting a given expression is easier for a model than finding the expression in running text, and that performance climbs with scale and with how the task is framed.

The generative side, rewriting figurative language rather than only labelling it, is far less developed, and the reason is measurement. A recent survey of figurative-language generation finds that methods have improved sharply with pre-trained models while evaluation has not, leaving evaluation as the dominant bottleneck: automatic metrics such as BLEU and BERTScore correlate only weakly with human judgement on figurative output [19]. This thesis sits on that generative side, and the measurement problem it names is the reason the evaluation here is built around human judgement (Chapter 4).

2.2 Easy-to-Read and Cognitive Accessibility

Easy-to-Read is a set of guidelines for writing text that more people can understand. Its readers include people with intellectual disabilities, people with low literacy, and people reading in a language that is not their first. The most widely used European guidelines, Inclusion Europe's *Information for All*, were written for people with intellectual disabilities and give plain rules for words, sentences, and layout [17]. In Spain the same practice is now a formal standard, UNE 153101:2018, which sets out how to produce and validate Easy-to-Read documents [3]. Easy-to-Read is therefore not an informal style but a codified accessibility requirement.

These guidelines treat figurative language as a barrier and tell writers to remove it. *Information for All* states the rule directly: "Do not use difficult ideas such as metaphors. A metaphor is a sentence that does not actually mean what it says", giving "it is raining cats and dogs" as the example to avoid [17]. The instruction is clear about the goal and silent about the method. It says that a metaphor or an idiom should not appear in an Easy-to-Read text, but it does not say how to turn one into a plain sentence that keeps the original meaning. That step is left to a human editor, who must understand the expression and rewrite it by hand. This is the gap the thesis addresses: the guideline names figurative language as something to remove, but neither it nor any tool says how to replace it.

Work at UPM has begun to automate Easy-to-Read adaptation one guideline at a time, building a separate module for each rule rather than one general rewriter. There is a module for morphological features, such as adverbs ending in *-mente* and superlatives [28], and a module for converting dialogue into a clearer form [7]. These modules belong to a research line whose support application, FACILE, assesses a document against the Easy-to-Read guidelines and then transforms it accordingly [27]. Each module automates one guideline and is checked with the readers it serves. This thesis is the figurative-language module of that pattern. It works in English, where the figurative-language resources are richest; porting it to Spanish, where the FACILE programme already operates, is left as future work (Section 8.7).

Easy-to-Read adaptation overlaps with automatic text simplification but is not the same task. Simplification research aims broadly to make sentences shorter or easier, often measured against general readability. Easy-to-Read adaptation serves a specific population and a specific set of guidelines, and removing figurative language is one of its explicit requirements rather than an optional improvement. This difference shapes how the output should be judged. A rewrite that keeps the meaning but stays hard to read has not met the Easy-to-Read goal, so the evaluation in this thesis scores simplicity as its own dimension alongside meaning preservation, a choice that becomes central to the results in Chapter 7.

2.3 Large Language Models & Agentic Systems

Large language models can be steered by their prompt alone, with no change to their weights. In zero-shot prompting the model is given only an instruction; in one- or few-shot prompting it is also shown one or a handful of worked examples and is expected to generalise from them [6]. The model's output can be constrained to a fixed format, such as a JSON object with named fields, so that a program can read the result reliably instead of parsing free text. A further technique, chain-of-thought (CoT) prompting, asks the model to write out intermediate reasoning steps before its final answer, which helps on tasks that need several steps of inference [31]. This thesis works in the zero- and one-shot setting throughout, because the replacement task has no training data to fine-tune on.

These techniques can be combined into systems that split a task into several steps, each handled by its own call to the model. For figurative language, this kind of decomposition has precedent. Theory-guided multi-step scaffolding, which walks the model through a linguistic procedure rather than asking for a verdict directly, improves metaphor reasoning over plain prompting [30]. A paraphrase-then-self-check strategy, in which the model first restates an idiom and then checks its own paraphrase, raises idiom understanding without any training [9]. Multi-stage pipelines have also been applied to idiom tasks with competitive results [4]. Together these results make the decomposition hypothesis of this thesis (H2) a motivated expectation rather than a hunch. A related pattern, augmenting the input with model-generated surrounding context, has likewise been shown to help metaphor detection [15].

That expectation is not a certainty. Supervised systems trained for the task remain strong: curriculum-style fine-tuning reaches state-of-the-art metaphor detection [18], so the assumption that a zero-/one-shot pipeline can match supervised performance (H1) cannot be taken for granted, and it is tested directly in Section 5.

A practical difference between a decomposed system and a single prompt is that the steps are visible. A multi-step pipeline records its intermediate state, what it detected, how it explained the expression, and what it produced, so a failure can be traced to the step that caused it. A single prompt that emits only a final answer leaves no such trace. This observability is a property of the design rather than a measured quantity, and it is examined qualitatively in the Discussion (Section 8.4).

2.4 Datasets

2.4.1 VU Amsterdam Metaphor Corpus (VUAMC)

The VU Amsterdam Metaphor Corpus [25] is a large, manually annotated corpus of English texts drawn from four genres: academic prose, news, fiction, and conversation. Metaphor annotation follows the MIP (Metaphor Identification Procedure) [26]: for each content word, annotators assess whether its contextual meaning contrasts with a more basic, concrete meaning from which the contextual meaning can be derived by comparison. Words meeting this criterion are labelled metaphorical.

The corpus is distributed as an XML file and contains 16,202 English sentences, each with per-token metaphoricity labels (binary: metaphorical / non-metaphorical). Crucially, the VUAMC provides *token-level* annotations rather than span- or sentence-level ones, making it well-suited for evaluating fine-grained detection but challenging for span-based models, since a single conceptual metaphor may involve multiple individually annotated tokens.

A custom Python client was developed to parse the XML, extract sentence-token structures, and derive character-offset spans from the token annotations. All 16,202 sentences are ingested into the experiment database. The VUAMC does *not* include gold literal paraphrases; replacement evaluation on this dataset therefore relies on detection metrics and human evaluation only.

2.4.2 SemEval-2022 Task 2: Multilingual Idiomaticity Detection and Sentence Embedding

SemEval-2022 Task 2 [29] is a shared task on idiom detection and interpretation in English, Portuguese, and Galician. This thesis uses the English (EN) partition of Subtask A and Subtask B:

- **Subtask A** provides sentences containing a pre-identified multi-word expression (MWE) and asks whether the MWE is used idiomatically in context (binary classification). Gold annotations include character-offset span boundaries for the MWE.
- **Subtask B** provides gold literal paraphrases of the full sentence for a subset of Subtask A examples, enabling evaluation of replacement quality.

The English partition contains 3,487 sentences after filtering and joining Tasks A and B. Gold paraphrases from Task B are stored alongside each example and used as references for

BLEU and BERTScore evaluation. Unlike the VUAMC, SemEval-2022 Task 2 is evaluated at the sentence and span level, with no per-token labels.

2.4.3 Other Figurative-Language Datasets

- MAGPIE [14]: idiom interpretation robustness (qualitative / stress testing)
- TroFi [5]: verb metaphor detection (focused evaluation)
- MOH / MOH-X [21]: verb metaphor detection (secondary validation)
- MetaNet: conceptual metaphor grounding (interpretation support)

2.5 Evaluation Methodology

Standard automatic metrics are commonly applied to generation and paraphrase tasks, but their suitability for figurative language is limited. This section outlines the key metrics used in this thesis and discusses their limitations in the context of figurative-to-literal adaptation.

2.5.1 Automatic Metrics

BLEU. BLEU measures n-gram overlap between a system output and one or more reference texts [23]. It is precision-oriented and applies a brevity penalty. For figurative language replacement, BLEU is severely misaligned: literal paraphrases of idiomatic expressions intentionally avoid lexical overlap with the gold reference, and multiple semantically equivalent paraphrases may share no surface n-grams. BLEU is retained here primarily as a baseline point of comparison, but its low values should not be interpreted as evidence of poor replacement quality.

SARI. SARI decomposes quality into Add, Delete, and Keep operations relative to the input and references, and was originally designed for text simplification [32]. Figurative-to-literal transformation closely resembles simplification (replacing opaque expressions with transparent ones), making SARI conceptually well-motivated. It is nevertheless not used in this thesis: its reference dependence makes it inapplicable to the metaphor track (the VUAMC has no gold paraphrases), and on the idiom track it would be redundant with the human evaluation, which is the load-bearing instrument for replacement quality.

BERTScore. BERTScore computes contextual token-embedding similarity between generated and reference texts using a pre-trained language model as its embedding backbone (here the default RoBERTa model that BERTScore ships with, not a model trained for this thesis), enabling soft alignment rather than exact string matching [33]. This makes it more robust to paraphrase variation than BLEU. However, high BERTScore does not guarantee correct interpretation: a plausible but semantically incorrect paraphrase may still achieve a high score. BERTScore is used as the primary automatic metric for replacement evaluation.

2.5.2 Human Evaluation

Automatic metrics alone are insufficient for assessing figurative-to-literal replacement quality. Recent work in sentence simplification has shown that most automatic metrics correlate only spuriously with human judgement on this class of task [2, 24], motivating a primary reliance on human evaluation. The protocol adopted is *Direct Assessment* [12, 13], applied to the three-axis rubric (grammaticality, meaning preservation, simplicity) established for sentence simplification by Alva-Manchego et al. [1]. Human evaluation is reserved for the open-ended task of replacement, where no single ground truth exists; detection has gold annotations and is evaluated only with the automatic metrics above. The replacement evaluation covers idioms only (SemEval Task B); metaphors are evaluated through detection metrics on detect-then-replace runs (Section 7). The full survey methodology (continuous-scale design, standardisation and aggregation, quality control, and three deliberate departures from the original DA specification) is the subject of Chapter 4; the survey instantiation (system conditions, source-sentence pool, respondent recruitment) is part of the RQ3 experimental setup (Section 7.5).

2.5.3 Multi-Layer Evaluation Framework

The evaluation protocol is structured as four complementary layers, inspired by the framework proposed in [27]:

1. **Detection accuracy**: span-level precision, recall, and F1 (Section 3.4);
2. **Semantic similarity**: BERTScore and BLEU (supporting/automatic);
3. **Interpretive correctness**: human evaluation of meaning fidelity (primary);
4. **Conceptual grounding**: correctness of source–target metaphor mappings and idiom sense disambiguation (not evaluated in this thesis; left to future work, Section 8.7).

2.6 Synthesis and Research Gap

The two bodies of work reviewed above, figurative-language processing and Easy-to-Read adaptation, do not meet. Research on figurative language is centred on detection: it builds systems that find and label idioms and metaphors, and even its generative branch stops at producing figurative text rather than removing it for a reader. Research on Easy-to-Read adaptation takes the opposite stance, rewriting text for accessibility, but it automates other guidelines and treats figurative language as something to be avoided rather than adapted. No prior work treats figurative-language adaptation as an Easy-to-Read guideline module, which is to say no one detects a figurative expression and then replaces it with an accessible literal form. That intersection is the gap this thesis fills.

The way it tries to fill it rests on a bet that the literature makes plausible but does not settle. Several studies suggest that decomposing a figurative-language task into explicit steps helps. Theory-guided multi-step scaffolding improves metaphor reasoning over plain prompting [30], and a paraphrase-then-self-check prompting strategy raises idiom understanding without extra training [9]; multi-stage pipelines have likewise been applied to idiom tasks with competitive results [4]. These gains are reported on large models, and whether the same decomposition still helps on a mid-size open-weights model that can be deployed in practice is exactly what this thesis tests in RQ2.

The second open question is how to judge the result. No automatic metric is trusted for figurative-language replacement: the generation survey names evaluation as the field's dominant bottleneck [19], and the simplification literature shows that common metrics correlate only spuriously with human judgement on this class of task [2, 24]. A study of figurative-to-literal rewriting therefore cannot lean on automatic scores alone; it has to be grounded in human judgement, and Chapter 4 builds that instrument. In short, the literature offers structure as a promising lever and warns that no metric can settle it; the next two chapters take both seriously.

3 System Design & Methodology

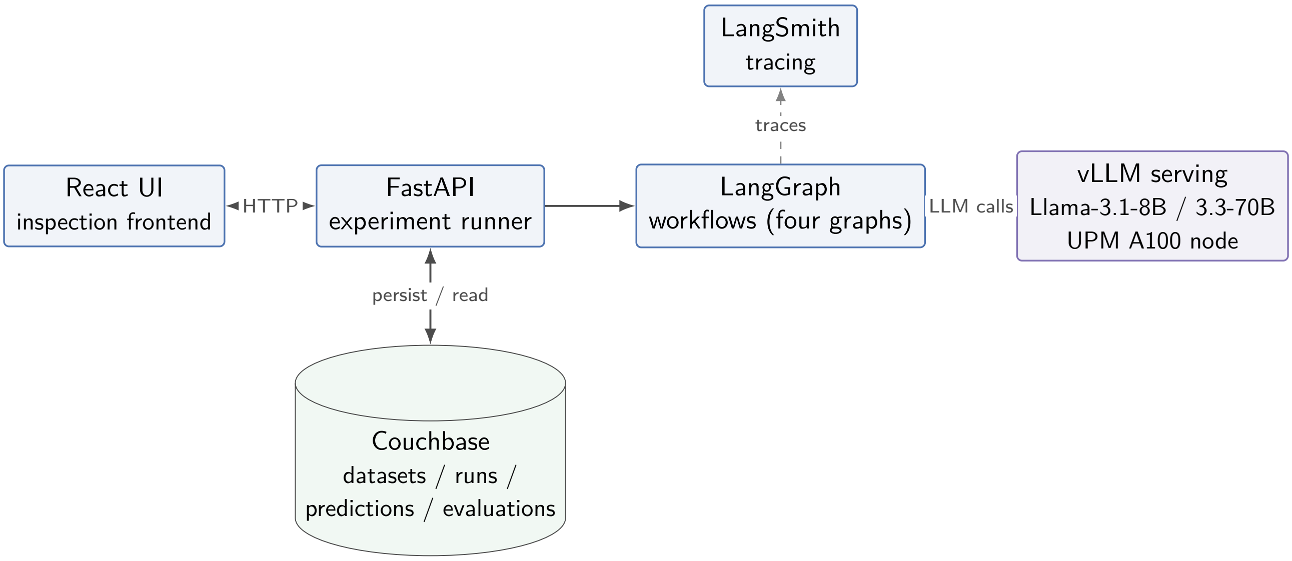

Figure 1: System architecture. The FastAPI experiment runner drives the four LangGraph workflows, which issue calls to the open-weights models served by vLLM on the UPM A100 node. Datasets, runs, predictions, and evaluations are persisted in Couchbase; every model call is traced to LangSmith (dashed) for the failure analysis in Section 8.4; and the React frontend reads runs back for inspection.

3.1 Problem Formulation

The core task addressed by this thesis is figurative language adaptation: given a raw English sentence that may contain figurative expressions (idioms or metaphors), produce a literal, meaning-preserving restatement alongside explicit detection annotations. More formally, the system accepts a sentence s and outputs:

- A set of detected figurative spans $\{(e_i, \text{start}_i, \text{end}_i)\}$, where each e_i is a verbatim copy of an expression from s with character-offset boundaries;
- A per-token binary label sequence $\mathbf{l} \in \{0,1\}^{|\mathcal{T}(s)|}$ over the whitespace-tokenised form of s, derived from the detected spans;
- A brief linguistic explanation of the figurative usage;
- Optionally, a literal paraphrase $\hat{s}$ of the full sentence, with all figurative expressions replaced by meaning-preserving literal alternatives.

Two task variants are supported: *detection* (spans, labels, and explanation only) and *detect-then-replace* (additionally producing $\hat{s}$). Both are evaluated over two phenomena independently: *metaphors*, benchmarked against the VU Amsterdam Metaphor Corpus [25], and *idioms*, benchmarked against SemEval-2022 Task 2 [29]. A notable challenge is that some metaphors encode meaning that resists full literalisation without loss; replacement decisions therefore require nuanced linguistic understanding beyond surface paraphrase.

3.2 Agentic Pipeline

The system is implemented as four LangGraph workflows, one per (phenomenon, task) combination:

- `metaphor_detection_graph`: detects metaphorical words or phrases, following VUAMC MIP (Metaphor Identification Procedure) guidelines;
- `idiom_detection_graph`: detects multi-word idiomatic expressions, following SemEval-2022 Task 2 annotation conventions;
- `metaphor_detect_then_replace_graph`: detects metaphors and produces a literal sentence paraphrase;
- `idiom_detect_then_replace_graph`: detects idioms and produces a literal sentence paraphrase.

Each workflow issues a single structured LLM call whose output must conform to the `TaskOutput` Pydantic schema:

- `detection.is_figurative` (boolean): whether the sentence contains any figurative expression;
- `detection.figurative_expressions` (list of strings): verbatim copies of each detected expression, preserving casing and punctuation as they appear in the input text;
- `detection.explanation` (string): brief linguistic reasoning for the detection decision;
- `replacement.literal_paraphrase` (optional string): the complete sentence rewritten with all figurative expressions replaced.

Structured output is enforced via LangChain's `with_structured_output()` interface, which uses JSON-schema-constrained decoding or function-calling depending on the model's capabilities. Outputs violating the schema trigger automatic retries.

Two deterministic post-processing steps follow the LLM call:

1. **Span extraction**: each verbatim expression in `figurative_expressions` is located in the input sentence by exact string search, with a case-insensitive fallback. Matched positions are converted to `Span(start, end)` objects with inclusive-start, exclusive-end character offsets. Overlapping spans are merged; all spans are clipped to the sentence boundary.
2. **Token label derivation**: a binary label is assigned to each whitespace-delimited token by testing whether any detected span overlaps the token's character range. This produces a label sequence directly comparable to VUAMC's token-level metaphor annotations.

The workflows above issue a single structured call. The system also implements the same task as an explicit *three-step pipeline*, in which detection, explanation, and transformation are three sequential LLM calls rather than one. The detection step commits to the figurative expressions; the explanation step states, for each expression, its literal meaning; and the transformation step rewrites the sentence using those meanings. Each step's output is kept in the workflow state, so the intermediate results are inspectable on their own rather than collapsed into one opaque response. This pipeline is the agentic condition used in the human evaluation (Section 7.5), and its contrast with the monolithic baseline (Section 3.5) is the designed experiment behind RQ2. Figure 2 shows the structure of the pipeline, and Figure 3 traces a real sentence through it.

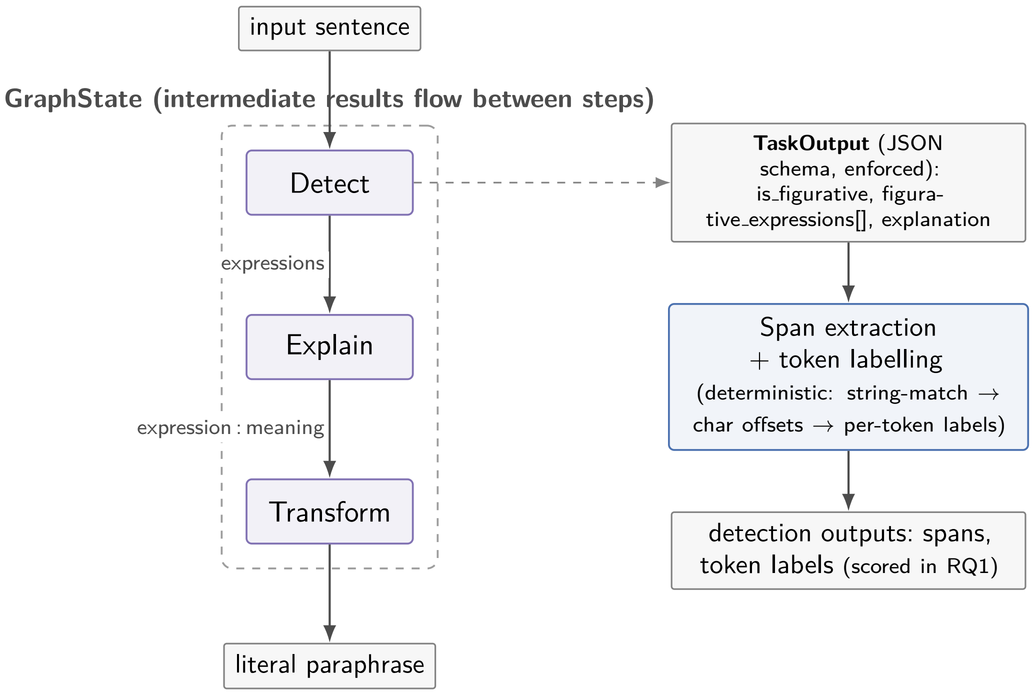

Figure 2: The three-step agentic pipeline as implemented. Detection, explanation, and transformation are three sequential LLM calls in a LangGraph workflow, with intermediate results carried in the graph state. The detection call returns a schema-constrained `TaskOutput` object (a schema-invalid response triggers an automatic retry); two deterministic post-processing steps then derive the character-offset spans and per-token labels scored in Section 5. The explanation step produces a literal meaning for each detected expression, and short-circuits to an unchanged sentence when nothing is detected; the transformation step rewrites the sentence using those meanings, and the per-expression explanations are persisted for the H4 analysis (Section 8.6).

3.3 Infrastructure

The system is built on the following components:

- **Backend**: FastAPI (Python 3.13). Experiment runs are launched asynchronously as background tasks, enabling concurrent execution.
- **LLM access**: a provider-dispatched client wrapping LangChain's `ChatOpenAI`. Model identifiers are routed at runtime: those prefixed with `vllm:` are served by the open-weights cluster (see below), and unprefixed identifiers pass through to OpenRouter for development-time smoke-testing. All experimental results reported in this thesis are produced by the open-weights cluster; OpenRouter is retained only as a development convenience for system-shakedown runs.
- **Open-weights serving**: vLLM (continuous batching, PagedAttention) on a UPM A100 GPU node, exposed to the experiment runner via a basic-auth-protected ngrok HTTPS tunnel. Two open-weights models are evaluated: `meta-llama/Meta-Llama-3.1-8B-Instruct` (FP16, deliverable system) and `casperhansen/llama-3.3-70b-instruct-awq` (AWQ-quantised, scale-comparator). The same OpenAI-compatible client code path serves both, so the only difference between conditions is the model identifier on the run document.
- **Persistence**: Couchbase 7.6 document store with four collections: `datasets` (gold-annotated examples; 16,202 VU Amsterdam + 3,487 SemEval = 19,689 total), `runs` (experiment metadata and status), `predictions` (per-example LLM outputs), and `evaluations`

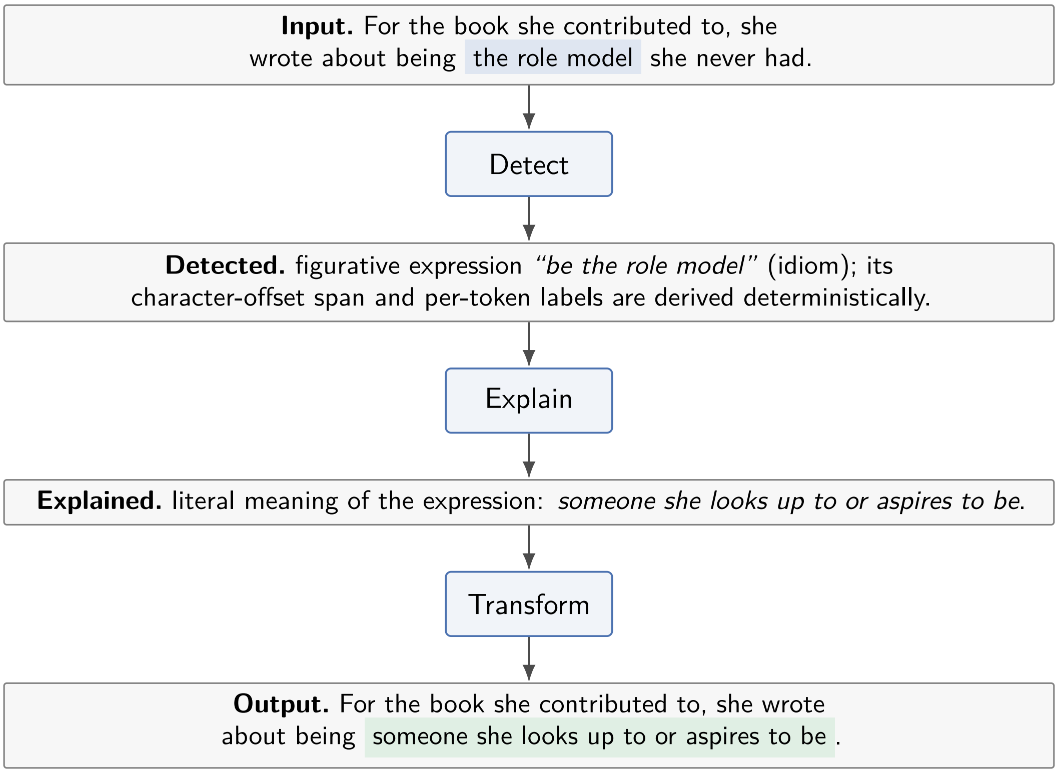

Figure 3: The three-step agentic pipeline traced on a real example from the survey pool (the 8B agentic system on SemEval source `src_02`). Detection commits to the figurative expression as a verbatim span; explanation states its literal meaning; transformation substitutes that meaning back into the sentence. Input and output are shown verbatim; the highlighted phrases mark the replaced expression. The detection output additionally passes through the two deterministic post-processing steps above before evaluation.

(computed metrics). Document keys follow a namespaced convention (e.g. `prediction::{run_id}::{example_id}`).

- **Prompt versioning**: each run records the full prompt text and its SHA-256 hash, enabling exact reproducibility and detection of prompt drift between runs. The full text of every prompt (detection, explanation, transformation, and the monolithic baseline) is reproduced in Appendix 10.2.
- **Observability**: LangSmith tracing is configured for all LLM invocations, providing post-hoc inspection of model inputs, outputs, token usage, and latency per step. Trace analysis is discussed qualitatively in Section 8.4.
- **Frontend**: a React/TypeScript single-page application with TanStack Query for data fetching, featuring run dashboards, example-level span visualisation with colour-coded gold/predicted overlays, literal replacement diff views, and multi-run metric comparison charts.

3.4 Evaluation Framework

Five evaluation metrics are computed automatically for each completed run:

- **Token F1**: binary precision, recall, and F1 aggregated across all token labels within a run, computed using scikit-learn's `precision_recall_fscore_support`. Applicable only when gold token-level annotations are available (VU Amsterdam).

- **Span F1**: macro-averaged precision, recall, and F1 over detected character-offset spans. A predicted span is a true positive if it overlaps any unmatched gold span with Intersection-over-Union (IoU) ≥ 0.5, enforcing one-to-one matching.

- **Sentence F1**: binary classification F1, where a sentence is labelled figurative if and only if at least one span is detected, for both gold and predicted.

- **BLEU**: sentence-level BLEU with NLTK smoothing (method 1) against the gold paraphrase from SemEval Task B. Applicable only to detect-then-replace runs on SemEval.

- **BERTScore** [33]: contextual embedding similarity (using BERTScore's default RoBERTa backbone) between predicted and gold replacement sentences, returning precision, recall, and F1. Applicable only to detect-then-replace runs on SemEval.

All metrics are persisted as `EvaluationData` Couchbase documents and are re-computable on demand via the `POST /runs/{run_id}/evaluate` endpoint. Section 2.5 discusses the appropriateness of these metrics for figurative language evaluation in more detail.

3.5 Baselines

- **Monolithic single-prompt**: a one-paragraph stub prompt that requests a literal rewrite of the input sentence with free-text output. No detection step, no explicit reasoning step, no structured output schema. This is the RQ2 control condition (Section 6) and one of the three human-evaluation survey conditions (Section 7.5).

- **Detection-only comparison**: detect-then-replace runs are compared against detection-only runs on the same examples to quantify the effect of the replacement step on detection quality (see Sections 5 and 7).

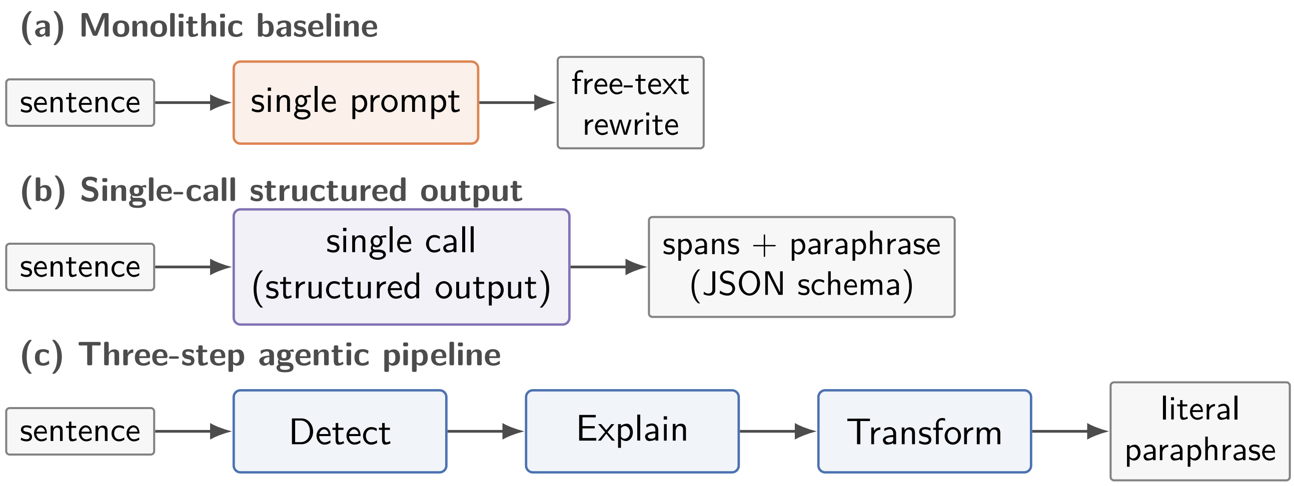

Figure 4: The three system architectures compared in this thesis. The monolithic baseline (a) produces only a free-text rewrite; the single-call structured-output workflow (b) commits to detected spans alongside the paraphrase in one call; the three-step agentic pipeline (c) separates detection, explanation, and transformation into sequential steps with inspectable intermediate state. Structure increases from (a) to (c); the comparison between them is the designed experiment behind RQ2.

4 Methodology: Human Evaluation Survey

The survey covers idiom replacement only: the VUAMC provides no gold paraphrases, and the three-part rubric below is well-defined for the figurative-to-literal mapping of idioms but breaks down for conventional metaphors that may require no rewrite at all (Section 7 develops this scoping decision).

This chapter specifies the human-evaluation protocol used to assess figurative-to-literal replacement quality for RQ3 (Section 7). The replacement task lacks a single ground truth: for an idiom such as *smoking gun*, valid literal paraphrases include *decisive evidence*, *conclusive proof*, or longer rewrites that preserve sentence meaning, none of which need share surface form with each other or with a gold reference. As discussed in Section 2.5, automatic metrics in this regime correlate only spuriously with human judgement [2, 24], so human evaluation is the primary signal for replacement.

The protocol adopted here is *Direct Assessment* (DA), the standard evaluation method used at the Conference on Machine Translation (WMT), introduced for segment-level machine translation (MT) by Graham et al. [12] and extended to document-level reliability by Graham et al. [13]. The three-part rubric layered on top of DA (grammaticality, meaning preservation, and simplicity) follows the sentence-simplification convention established by Alva-Manchego et al. [1]. Section 4.1 describes the protocol; Section 4.2 documents three deliberate departures from the original Graham specification, each justified by a difference between the machine-translation setting and the figurative-language replacement setting. The instantiation of the protocol (specific systems compared, source-sentence pool, item counts, and respondent recruitment) is described in the experimental setup of RQ3 (Section 7.5).

4.1 Direct Assessment Protocol

4.1.1 Continuous-scale Rating

For each evaluated item, the annotator rates the system's literal paraphrase on three independent 0–100 continuous sliders, one per rubric dimension (defined below). The continuous scale is the foundational design choice of the protocol: as Graham et al. [12] demonstrate, replacing a discrete (e.g., 1–5) Likert-style scale with a 0–100 continuous slider unlocks two assessor-intrinsic quality-control techniques that Likert scales cannot support, raises intra-annotator κ by up to +0.101, and raises inter-annotator κ by up to +0.144 when combined with per-annotator standardisation. Crowd-sourced annotators on platforms such as Mechanical Turk reach acceptable agreement under this design without expert review.

4.1.2 The Three Rubric Dimensions

The three dimensions are rated independently:

- **Grammaticality**: is the output well-formed English?
- **Meaning preservation**: does it convey the meaning of the original sentence?
- **Simplicity**: is the output more accessible than the figurative source?

Annotators may optionally provide an alternative paraphrase, retained as qualitative data for the discussion section. Annotators are blind to the system-of-origin; per-condition identifiers live only in analysis-side metadata.

4.1.3 Standardisation and Aggregation

Following Graham et al. [12], raw scores are converted to z-scores per annotator (using each annotator's own mean and standard deviation across all rated items) before aggregation. This corrects for individual scoring tendencies ("score-low" versus "score-high" annotators rating the same outputs) and is the property that makes the continuous scale recover meaningful inter-annotator agreement where Likert-style scales do not.

Aggregation follows the micro-then-macro pattern of [13]: the mean standardised score is computed per item across its repeat ratings, then per condition across items. Per-item scores are reported as standardised mean $\pm$ 95% confidence interval, with a composite of the three rubric dimensions as a summary statistic. Confidence intervals use the Student-t distribution over the standardised ratings; given the small per-item repeat counts (Section 4.2.2), they are reported as indicative rather than inferential.

4.1.4 Quality Control

Two assessor-intrinsic checks are interleaved into each respondent's set:

- **Bad-reference items**: deliberately degraded versions of system outputs, constructed by lexical substitution or noun-phrase swap that breaks meaning preservation while keeping the sentence superficially fluent. The annotator should rate the bad reference lower than the corresponding genuine system output on the targeted rubric dimension.
- **Repeat-pair items**: duplicate items presented twice in the same respondent's set, separated by other items. The absolute score difference quantifies intra-annotator consistency.

Both techniques are introduced in [12]; the canonical crowd-sourcing implementation (using bad references inside larger HITs, scored by a per-worker difference-of-means significance test, and retaining only workers with $p < 0.05$) is described in [13]. The thesis uses fixed-threshold filters in place of the per-worker significance test, for reasons documented in Section 4.2.3.

4.2 Departures from the Original Protocol

The DA protocol as described above follows Graham et al. [12, 13] closely. Three design decisions diverge from the original specification, each motivated by a feature of the figurative-language replacement task that does not hold in the machine-translation setting Graham analyses.

4.2.1 Third Dimension versus Fluency Omission

Graham et al. [13], Section 3.3, argue that DA fluency assessment can be safely dropped when resources are constrained: an earlier study [11] found no evidence of reference bias in DA adequacy, removing the original methodological motivation for a separate fluency dimension; the freed budget is better spent expanding the test set on the adequacy dimension.

The thesis retains a third dimension (*grammaticality*) in deliberate tension with this argument, on three grounds. First, the task is text simplification, not machine translation: the simplification literature [1] converged independently on a three-axis rubric (grammaticality, meaning preservation, simplicity) because the additional dimension carries information that adequacy alone does not capture. Second, small open-weights LLMs at the scale used here (Llama-3.1-8B and Llama-3.3-70B-Instruct-AWQ; see Section 7.5) produce qualitatively different error modes from professional MT systems: agreement and tense errors, dropped articles, and clipped continuations are visible in pilot outputs and would be invisible to a meaning-only rubric. Third, the budget freed by dropping the third axis would translate to perhaps four to five additional items in our 81-item, 22-respondent design, a marginal reliability gain that does not compensate for losing a diagnostic axis.

4.2.2 Reliability Bound and Exploratory Reporting

Graham et al. [13] establish, by self-replication on Mechanical Turk over the WMT-16 EN–ES document-level QE test set, that DA scores reach $r = 0.901$ between two independent data-collection runs at ≥ 27 repeat assessments per document (averaging ≈ 107 segment-level judgements per document). The corresponding segment-level threshold from [12] is ≥ 15 repeat assessments. These thresholds define the reliability frontier above which DA can be used as a substitute for an expert gold standard.

The thesis survey ships with a target of ≥ 3 ratings per item, well below this frontier. This is a deliberate choice driven by the scale of the deliverable: 81 items at 20–25 respondents (16 rated items + 4 quality-control items per respondent) is at the upper end of what a master's-thesis recruitment effort can sustain, and tripling the respondent count to approach Graham's threshold is not feasible. The methodological consequence is that per-condition scores are not used to make reliability-bounded claims about absolute system quality. Instead, the survey is treated as exploratory: results are reported per condition with the standardised means defined above, and pairwise condition comparisons, paired by source sentence, are tested with the paired t-test and the Wilcoxon signed-rank test on the per-source mean standardised scores. The survey supports the qualitative claims of RQ2 (decomposition vs. monolithic) and RQ1 (within-family scale comparison) at a directional level; it does not establish absolute quality estimates.

4.2.3 Quality-control Filter Design

Graham et al. [13] embed bad-reference items inside 100-translation HITs and apply a per-worker difference-of-means significance test, retaining only workers with $p < 0.05$. The test has good power at the original HIT size: with ~ 100 items per worker including a balanced bad-reference contingent, a worker who fails to discriminate degraded from genuine outputs is reliably flagged.

The thesis design has 20 items per respondent (16 rated + 4 QC), of which only two are bad references and two are repeat pairs. With $n = 2$ per QC type, the per-respondent significance test has effectively no power: a respondent could rate both bad references higher than their corresponding genuine outputs and still fail to reach $p < 0.05$ under the t-test. Fixed-threshold filters are used instead, applied in three rules. (a) *Completion*: respondents who fully rate fewer than 10 of their 20 items are excluded. (b) *Repeat-pair consistency*: respondents whose mean absolute first-versus-second-showing difference across their repeat pairs exceeds 25 points (on the 0–100 scale) are excluded. (c) *Bad-reference discrimination*: each bad-reference's *meaning preservation* score is compared against the respondent's own mean meaning score across their genuine items; a respondent both of whose bad references score at or above that mean is flagged, and headline results are reported with and without flagged respondents. The bad-reference rule targets the meaning dimension because the degradations are constructed to break meaning preservation specifically while remaining superficially fluent; comparing against the respondent's *own* mean rather than a global threshold preserves the assessor-intrinsic character of the original technique under per-annotator score variation. Threshold values are fixed in advance and reported with the survey; they are not tuned post-hoc against the rated data. The construction recipe for the bad-reference and repeat-pair items is detailed in Appendix 10.1.4.

5 RQ1: Detection & Interpretation

To what extent can a zero-/one-shot agentic LLM pipeline accurately detect and interpret idiomatic and metaphorical expressions in context?

5.1 Experimental Setup

All detection experiments use the following configuration:

- **Models**: two open-weights LLMs served via vLLM on the UPM A100 cluster: `Llama-3.1-8B-Instruct` (the deliverable system) and `Llama-3.3-70B-Instruct-AWQ` (a same-family scale comparator). Both are accessed through the same OpenAI-compatible interface via the experiment runner's `vllm:` dispatch prefix.
- **Temperature**: 0 (greedy decoding, for reproducibility);
- **Prompt version**: v1 (zero-shot, no few-shot examples);
- **Evaluation set**: the full evaluation sets (16,202 VUAMC metaphor sentences; 3,487 SemEval idiom items).

The exact model identifiers, serving configuration, and a complete index of the run identifiers behind every result reported in this thesis are collected in Appendix 10.4.

For metaphor detection, the prompt instructs the model to follow the MIP procedure: identify each word whose contextual meaning contrasts with its most basic, concrete meaning, and return those words verbatim as `figurative_expressions`. For idiom detection, the prompt instructs the model to identify multi-word expressions whose combined meaning cannot be derived compositionally.

In both cases, the LLM output is post-processed to extract character-offset spans and derive per-token binary labels (Section 3.4). These are compared against gold annotations from the respective datasets.

5.2 Results

Table 1 reports detection-only results on both Llama-3.1-8B-Instruct and Llama-3.3-70B-Instruct-AWQ, prompt v1, $T = 0$, on the full evaluation sets (16,202 VUAMC metaphor sentences and 3,487 SemEval idiom items). The within-family scale dimension and the per-dataset detection profile are read off the same table.

5.3 Analysis

What these detection scores measure. The absolute figures carry a ceiling set by the gold standards rather than by the models alone, and the comparisons in this section are read with that in mind. The VUAMC follows the MIPVU procedure, which marks every metaphorically-used word, including the conventional and grammatical metaphors that saturate ordinary English (for example, the preposition *in* of *in trouble*, or the verb *give* of *give a talk*, both marked metaphorical by MIPVU yet invisible to an ordinary reader). The systems flag the metaphors a reader would notice and leave the conventional ones unmarked, so token recall against an exhaustive gold is low by construction, and precision exceeds recall throughout. The score measures agreement with MIPVU's full annotation rather than detection of the figurative

Dataset	Model	F1-sent	F1-tok	P-tok	R-tok	F1-span	Δ at scale
VU Amsterdam	8B-Instruct	0.254	0.206	0.238	0.183	0.033	n/a
VU Amsterdam	70B-Instruct-AWQ	**0.302**	**0.265**	**0.355**	**0.212**	**0.064**	+4.8pp F1-sent
SemEval-2022	8B-Instruct	0.333	n/a	n/a	n/a	0.204	n/a
SemEval-2022	70B-Instruct-AWQ	**0.529**	n/a	n/a	n/a	**0.337**	+19.6pp F1-sent

Table 1: Zero-shot detection results across model scales, on the full evaluation sets ($T = 0$, prompt v1). Token-level metrics are not reported for SemEval, whose gold annotations are span- and sentence-level only. Scaling from 8B to 70B-AWQ lifts idiom sentence F1 by +19.6pp (0.333 → 0.529) but metaphor sentence F1 by only +4.8pp (0.254 → 0.302); on metaphor the gain is concentrated in token precision (0.238 → 0.355) while recall barely moves (0.183 → 0.212). Runs: VU 8B `2d322ad0`, VU 70B `d47a01fb`, SemEval 8B `8141d880`, SemEval 70B `7cbef03f`.

language that actually needs rewriting, since conventional metaphors usually need no rewrite for an Easy-to-Read reader (Chapter 8). A second effect is granularity: span F1, which counts a predicted span as correct only when its overlap with the gold span (Intersection-over-Union, IoU) reaches 0.5, penalises the boundary mismatch between the models' verbatim phrase outputs and the token-level gold, which is why span F1 sits well below token and sentence F1 for metaphor. With the zero-shot setting on top, the absolute scores understate practical competence and are not comparable to the supervised systems of Table 2. The conclusions drawn here are therefore relative: they compare scale, prompt, and phenomenon on the same metric and data, where this ceiling cancels out.

Idiom detection exceeds metaphor detection at both scales. On the full sets, idioms are detected more accurately than metaphors on both metrics: idiom sentence F1 is 0.333 (8B) and 0.529 (70B) against metaphor's 0.254 and 0.302, and idiom span F1 is 0.204 and 0.337 against metaphor's 0.033 and 0.064. This is consistent with H3, which holds idioms to be the more tractable phenomenon, and with the metric ceiling noted above: metaphor detection is scored against MIPVU's exhaustive token-level gold, so the systems' recall is bounded by how much of that exhaustive marking they reproduce, whereas idiom detection targets a single multi-word expression per item.

Scale lifts idiom detection but barely metaphor detection. The step from 8B to 70B is not a uniform gain. It lifts idiom sentence F1 by 19.6 percentage points (0.333 → 0.529) and idiom span F1 by 13.3 points (0.204 → 0.337), a substantial improvement. On metaphor the same step gains only 4.8 points of sentence F1 (0.254 → 0.302). The reason is visible in the token-level split: the 70B's token precision rises sharply (0.238 → 0.355) while its recall barely moves (0.183 → 0.212), so the larger model is more selective about which metaphors it flags but recovers few that the 8B missed, and against an exhaustive gold that selectivity buys little. Scale therefore helps detection where the target is a discrete expression (idioms) far more than where the gold demands exhaustive marking (metaphor); the gain is real but phenomenon-dependent, not the uniform lift a single headline figure would suggest.

The detect-then-replace prompt lifts detection on 8B but *costs* F1 on 70B, on both phenomena. Adding the auxiliary replacement field to the detection prompt (the detect-then-replace, or dtr, setting) raises 8B detection but lowers 70B detection: the same change with opposite signs across scale. On the 8B, sentence F1 rises from its detection-only baseline by +25pp on SemEval idiom (0.333 → 0.584) and +10pp on VU metaphor (0.254

$\rightarrow$ 0.352); forcing the model to commit detected spans alongside a paraphrase scaffolds the decision at small scale. On the 70B, the same prompt *lowers* sentence F1 by −3pp on SemEval idiom (0.529 $\rightarrow$ 0.496) and −11pp on VU metaphor (0.302 $\rightarrow$ 0.190), the larger model being calibrated enough that the schema constraint distorts rather than scaffolds, over-committing to "no figurative content" on borderline items (the 70B metaphor dtr also produced more malformed outputs, which compounds the drop). The sign flip, $+10/+25$pp at 8B versus $-3/-11$pp at 70B, is the structure-scale interaction developed for RQ2 (Section 6) and the Discussion (Chapter 8): a prompt change validated on a small model can reverse at scale. The runs behind these figures are listed in the appendix run index (Appendix 10.4).

5.4 Comparison to Supervised Benchmarks

Task	Supervised system	Supervised F1	This work	Comparability
VU metaphor	DM-ENS [22]	0.766 (token)	0.302 (sent.)	token- vs. sentence-level; supervised vs. zero-shot
SemEval idiom	Top system [29]	≈0.90 (macro)	0.529 (sent.)	macro vs. sentence F1; supervised fine-tuned vs. zero-shot LLM

Table 2: Indicative comparison of this work's strongest zero-shot detection against representative supervised systems. The numbers are not directly comparable: metrics, test sets, and training regimes differ (final column), so the comparison shows direction, not rank. Supervised figures are token-level metaphoric-class F1 on the VUA all-content-words split (DM-ENS, the 2020 shared-task winner, as analysed in [22]) and macro F1 on the English SemEval-2022 Task 2 Subtask A zero-shot track.

Zero-shot detection is best read against what systems trained for these tasks achieve. Table 2 places this work's strongest detection numbers next to representative supervised results. The comparison is indicative only and is not a leaderboard ranking: the figures here come from the full evaluation sets at temperature zero, scored differently and on different test sets from the supervised work, so the columns align in task but not in evaluation setup.

On both tasks the supervised systems are clearly ahead. For metaphor, the best system in the analysis of [22] reaches a token-level F1 of 0.766 on the standard VUA content-word split, against this work's sentence-level F1 of 0.302 at 70B. These are not the same measurement (token against sentence, supervised against zero-shot), but the direction is not in doubt: a model trained on the VUAMC localises metaphor more accurately than a zero-shot prompt. For idioms, the top system on SemEval-2022 Task 2 Subtask A reaches a macro F1 near 0.90 even in the harder zero-shot track [29], where the system sees no training examples of the test idioms; this work's 70B sentence F1 is 0.529. Here the unit is the same, sentence-level binary classification, so the gap is more directly meaningful, and it is wide.

One point of terminology needs care. The shared task's zero-shot and one-shot tracks still use fine-tuned supervised systems; they differ only in whether examples of the specific test idioms are provided. This work's zero-shot setting uses no task-specific training at all, which is a stronger sense of the term, so the comparison understates how much supervision the SemEval systems received.

Together these results support a partial reading of H1: a zero-/one-shot pipeline detects figurative language competently but does not match systems trained for the task. Supervised methods are also still improving. Jia et al. [18] reach state-of-the-art metaphor detection by fine-tuning with curriculum-style data augmentation (F1 0.866 on MOH-X and 0.736 on TroFi), so the premise behind H1, that a general-purpose model can rival task-trained ones, cannot be assumed.

A caveat cuts the other way. Neidlein et al. [22] show that high supervised metaphor scores partly reflect the disambiguation of conventionalised metaphoric word senses rather than a general grasp of metaphor: the systems "excel on frequently seen word types with conventionalised metaphoric meanings, which is more akin to word sense disambiguation". The measured gap may therefore overstate the difference in competence on novel metaphor, where neither approach is strong.

5.5 Qualitative Examples

The aggregate detection scores above combine three recurring behaviours, shown here on the SemEval idiom pool. Gold spans are the dataset annotation; the highlighted spans are what the 8B deliverable predicted.

Clean detection. Where the expression is a fixed, well-known idiom, the system marks the gold span exactly. For the source *"his tribute menu led off with Le Grande Fromage, French for the big cheese, as his staff called him"*, both the 8B and 70B systems return the gold span, and the sentence-level and span-level judgements agree.

Why span F1 trails sentence F1. A sentence counts as detected if the system finds any figurative span, but span F1 also requires the right boundaries and the right number of expressions. In *"Gastineau is a fashion plate of boho chic mixed with luxury"*, the 8B system detects fashion plate but misses the second idiom boho chic. The sentence is a true positive, yet only one of the two gold spans is recovered, so span recall falls while sentence accuracy does not. This pattern, the more obvious idiom caught and a co-occurring one missed, is one reason span F1 sits well below sentence F1 in the scores above, alongside the boundary-offset mismatches discussed earlier.

Scale lifts detection. Some idioms are missed by the 8B model and recovered by the 70B. In *"Carissa said of Matteo's baby blues, which he appeared to inherit from his dad"*, the 8B system detects no figurative expression, while the 70B system marks the idiom. Cases like this make up the within-family detection gain reported earlier in this chapter: the larger model finds figurative spans the smaller one does not. These gains are confined to detection; whether the agentic structure adds anything at the replacement step is the question taken up in the next chapter.

6 RQ2: Agentic Decomposition

Does an explicit agentic decomposition (detection → explanation → transformation) improve figurative-language handling compared to monolithic prompt-based LLM approaches?

6.1 Experimental Setup

The RQ2 controlled comparison pairs two task variants on identical inputs and identical model:

- **Agentic 3-step pipeline** (this thesis's deliverable): three sequential LLM calls that (1) detect each figurative expression verbatim, (2) explain its meaning in context, and (3) transform the sentence with all figurative parts replaced, each step with structured output and inspectable intermediate state.
- **Monolithic single-prompt** (the control condition): a stub prompt asking only for a literal rewrite of the input sentence, with free-text output. No detection step, no explicit reasoning step, no structured output schema.

Both variants run on Llama-3.1-8B-Instruct, $T = 0$, on the SemEval-2022 Task A/B idiom subset. The monolithic condition uses a one-paragraph stub.

6.2 Automatic-Metric Comparison

The agentic and monolithic conditions cannot be separated by automatic metrics, for two structural reasons, and this is why the human evaluation carries the RQ2 verdict rather than supporting it.

First, the conditions are not comparable on detection. The monolithic condition emits a free-text rewrite with no detection structure, so detection metrics (sentence and span F1) are undefined for it: there is nothing to score. Detection is in any case not where decomposition contributes, as Section 6.4 shows.

Second, the replacement side has no adequate reference. The task admits many valid literal paraphrases that share no surface form, and SemEval-2022 Task A supplies no gold paraphrase at all. The Task B paraphrase that can be matched to a source through its multiword expression exists for only 9 of the 27 pool sources, and where it exists it is a different sentence that happens to contain the same idiom rather than a literal rewrite of the source. Reference-based scores computed against it therefore measure cross-sentence lexical overlap, not replacement quality, and cannot decide between the two systems. This is consistent with the broader result that BLEU and BERTScore correlate only spuriously with human judgement of meaning preservation and simplicity in figurative-to-literal tasks [2, 24]. The thesis was built around this limitation: the Direct Assessment survey (Section 7.5) is the load-bearing instrument for RQ2, not the automatic metrics.

Failure modes are visible to qualitative inspection. Spot-checking the survey-pool items shows characteristic differences between the conditions. Monolithic outputs frequently produce *whole-sentence rewrites* in which non-figurative phrases are also paraphrased. For the source "*Long story short, I opted for the Letsfit White Noise Machine because it has excellent reviews on Amazon, 14 sounds, a handy nightlight, and it only costs $20*", for instance, the monolithic system rewrote "opted for" → "chose" and "handy nightlight" → "a light that is easy to use", whereas the agentic system left non-figurative content unchanged and only substituted "Long story short” → “To summarize”. The agentic structured output anchors the rewrite to the detected span; the monolithic system has no such anchor and generalises freely. Whether the rubric rewards this discipline (under *meaning preservation*) or penalises it (under *simplicity*, where more aggressive simplification scores higher) is the empirical question the survey resolves.

Hypothesis status. Hypothesis H2 (decomposition > monolithic) is therefore neither supported nor refuted by automatic metrics on this evaluation. It stands or falls on the human ratings across the three rubric dimensions, presented next.

6.3 Human-Evaluation Results

Dimension	**Agentic**	**Monolithic**	Δz	t p	**Wilc.** p
Grammaticality	83.1	85.4	−0.086	0.450	0.509
Meaning preservation	81.6	72.6	+0.309	0.058	0.082
Simplicity	60.6	64.1	−0.091	0.611	0.485
Composite			+0.044	0.722	0.768

Table 3: Agentic vs. monolithic (both Llama-3.1-8B) on the human-evaluation survey. The Agentic and Monolithic columns are raw 0–100 means per dimension; Δz is the paired difference in per-annotator-standardised scores (agentic − monolithic, paired by source sentence, $n = 27$ sources), tested with the paired t-test and the Wilcoxon (Wilc.) signed-rank test. Composite is the mean of the three standardised dimensions (agentic $z = 0.123$, monolithic $z = 0.125$); raw means are not defined for it.

The three open-weights conditions are reported in full in Section 7.6; this section isolates the RQ2 cut, the paired comparison of the agentic pipeline against the monolithic single-prompt at 8B (Table 3). On meaning preservation the agentic system is rated higher, 81.6 against 72.6 on the raw 0–100 scale, a nine-point gap that remains after per-annotator standardisation and pairing by source ($n = 27$). This is the chapter’s central result, and it is stated with the care its statistics require: the agentic decomposition shows a consistent directional advantage on meaning preservation (+0.31 standardised units, p = 0.06-0.08) that approaches but does not reach conventional significance (the usual $p < 0.05$ threshold); we interpret this as suggestive support for H2, strengthened by convergent qualitative evidence of span-anchored rewriting, rather than as a confirmed effect.

No overall-superiority claim. The advantage does not extend to the composite. The mean of the three standardised dimensions is tied, $z = 0.123$ for the agentic system against 0.125 for the monolithic one, a difference that is negligible and far from significant (Wilcoxon $p = 0.77$). On the other two dimensions the monolithic condition is marginally ahead, on grammaticality (85.4 against 83.1) and on simplicity (64.1 against 60.6), though neither difference approaches significance. The simplicity gap is the more interpretable of the two and points back to the span-anchoring failure mode shown earlier in this chapter, where on the “Long story short” example the monolithic system rewrote several non-figurative phrases along with the idiom: its free whole-sentence paraphrase reads as simpler even as it drifts from the source, while the agentic system, anchored to the detected span, changes less and so preserves meaning more faithfully while reading as marginally less simple. The two conditions trade meaning fidelity against aggressive simplification, in opposite directions.

The trade favours decomposition for Easy-to-Read adaptation. The two error types do not harm the target readership equally. A replacement that quietly alters the meaning of a sentence is a serious failure for a reader who depends on the adaptation, whereas a replacement that is faithful but only moderately simpler is merely less helpful. A system that secures meaning fidelity at the cost of a few points of perceived simplicity is therefore better aligned with the goal than one that simplifies more aggressively at the risk of meaning. This is what the structural lever buys: an advantage that is real but narrow, confined to the single dimension Easy-to-Read adaptation can least afford to lose, and visible only to human raters because no automatic metric in this study could separate the conditions. H2 is supported in the qualified, dimension-specific sense the data warrant, not as a blanket claim of superiority, and the advantage sits on the replacement side of the task, not in detection, as the next section shows.

6.4 Effect of Decomposition on Detection

On the full evaluation sets, detection F1 is near-identical under the standalone detection-only prompt and the 3-step pipeline across both scales and both phenomena: VU metaphor 0.254/0.254 at 8B and 0.302/0.302 at 70B-AWQ; SemEval idiom 0.333/0.332 at 8B and 0.529/0.529 at 70B. All seven token-, span-, and sentence-level detection metrics match across runs to within rounding; the only residual difference is on SemEval 8B (sentence F1 0.333 vs. 0.332), traceable to a slightly different set of structured-output parse failures rather than to any difference in detection behaviour. The match is exact by construction up to those failures rather than coincidental: the pipeline's detection step issues the same standalone detection prompt at temperature 0, so on every successfully-parsed example its greedy decoding returns byte-identical output and therefore identical labels. The value of the comparison is that it rules out any hidden divergence (sample shift, prompt drift) and confirms that the pipeline's detection step gets essentially the F1 the detection prompt would have gotten alone, regardless of scale: whatever the pipeline architecture buys, it is not detection sharpness. This also attributes the single-call CoT's detection lift over both of these conditions to the structured-output schema's commitment scaffold rather than to chain-of-thought reasoning content per se, and it implies that decomposition's value, where it exists, must be sought on the replacement side: which is precisely where the human evaluation (Section 6.3) locates it.

6.5 Further Comparisons

The current RQ2 evaluation contrasts only two conditions (full agentic vs. monolithic). Two further comparisons are planned and discussed as future work in Chapter 8:

- **No-explanation step**: agentic structured output without the chain-of-thought reasoning field, isolating the role of explicit reasoning;
- **No-self-verification**: agentic pipeline without a final consistency check between the detected span and the produced replacement.

These comparisons require minor schema changes on the existing LangGraph workflow and are deferred to give the human-evaluation survey priority.

6.6 Failure Traceability

A secondary benefit of the agentic decomposition is that each error mode is attributable to a specific intermediate step. The structured output records the detected expression, the explanation of its meaning, and the produced replacement; failures localise to one of these. In the monolithic condition, by contrast, errors are “the rewrite is wrong” with no further structure. This argument is developed in the observability discussion of Chapter 8.

7 RQ3: Replacement Quality

How effectively can figurative expressions be replaced with literal, meaning-preserving alternatives?

7.1 Experimental Setup

Replacement experiments use the *detect-then-replace* task variant: the model detects figurative spans and in the same call produces a literal paraphrase of the full sentence. All experiments use the open-weights cluster (`meta-llama/Meta-Llama-3.1-8B-Instruct`, FP16; and `casperhansen/llama-3.3-70b-instruct-awq` as a scale comparator), temperature 0. Two prompt versions are evaluated:

- **Prompt v1**: a minimal stub instructing the model to identify figurative expressions and provide a literal paraphrase (three lines of instruction).
- **Prompt v2**: a chain-of-thought prompt instructing the model to (1) identify each expression, (2) reason about its meaning in context, and (3) rewrite the full sentence with all figurative parts replaced, keeping non-figurative content unchanged. Includes one illustrative example.

7.2 Idiom Replacement (SemEval-2022)

SemEval-2022 Task B provides gold literal paraphrases for a subset of Task A sentences, enabling automatic evaluation of replacement quality using BLEU and BERTScore. Table 4 reports detect-then-replace results across two prompts (v1 stub, v2 chain-of-thought) and two architectures (single-call structured output, 3-step pipeline) on both Llama-3.1-8B-Instruct and Llama-3.3-70B-Instruct-AWQ ($T = 0$). The two metric families rest on different bases, which the table separates: the detection metrics (F1-sent, F1-span) are computed on the full evaluation set, while the reference-based replacement metrics (BLEU, BERTScore) are computed on the SemEval Task B gold-paraphrase items within the deterministic 200-example slice, the only items for which a gold literal paraphrase exists. The replacement metrics are not refreshed to the full set: gold-paraphrase coverage over the full 3,487 items is too sparse to compute them reliably (Section 6).

The chain-of-thought prompt gives no measurable benefit on Llama-3.1-8B. On the gold-paraphrase subset, the v1 (minimal) and v2 (chain-of-thought) prompts produce statistically indistinguishable replacement outputs on 8B: BLEU 0.397 vs. 0.400 (+0.3pp) and BERTScore F1 0.914 vs. 0.915 (+0.1pp). The most likely explanation is that the Llama Instruct chat template already applies similar reasoning scaffolding by default, so spelling the identify–reason–rewrite steps out in the prompt adds little on this model. The thesis uses the v2 prompt throughout but does not claim a prompt-level gain from it.

The pipeline architecture's BLEU advantage shrinks to zero at scale. On 8B the 3-step pipeline beats single-call CoT v2 by +3pp BLEU (0.430 vs. 0.400). On 70B the same comparison flips slightly to a −1pp difference (0.414 vs. 0.423), within sample noise. BERTScore F1 is tied at ∼0.915 in all four cells. The pipeline's small-scale BLEU advantage is therefore concentrated at 8B and partially attributable to a known artefact: when the pipeline's detection step finds nothing, the transform step short-circuits and returns the input sentence unchanged;

Architecture	Prompt	Model	F1-sent	F1-span	BLEU	BERTScore F1
Single-call CoT	v1 (stub)	8B	—	—	0.397	0.914
Single-call CoT	v2 (chain-of-thought)	8B	0.584	0.259	0.400	0.915
Single-call CoT	v2	70B	0.496	0.272	0.423	0.915
3-step pipeline	n/a	8B	0.332	0.204	0.430	0.916
3-step pipeline	n/a	70B	0.529	0.336	0.414	0.915

Table 4: Idiom detect-then-replace on SemEval-2022 Task A/B, $T = 0$. Single-call CoT runs use the structured-output schema requiring detection + replacement in one inference; 3-step pipeline runs separate detection, explanation, and transformation into three sequential LLM calls. **Detection metrics** (F1-sent, F1-span) are computed on the full evaluation set (runs: 8B v2 `d3a7a15c`, 70B v2 `162b7339`, 8B pipeline `b9f31108`, 70B pipeline `4bacbfb2`); the v1 stub was not rerun for detection at full scale (—), as the thesis uses the v2 prompt throughout. **Replacement metrics** (BLEU, BERTScore) are computed on the Task B gold-paraphrase items within the deterministic 200-example slice, the only items with a gold literal paraphrase (runs: 8B v1 `b2b174ea`, 8B v2 `d6bc60b0`, 70B v2 `92941662`, 8B pipeline `3551e201`, 70B pipeline `56f19b8c`). No condition is highlighted: the differences across conditions are within sample noise, and BERTScore is tied at ∼0.915.

the input shares roughly 85% of its tokens with the gold paraphrase, so those examples score artificially high on BLEU. This artefact contributes more to the 8B aggregate (which has lower detection recall, so more fall-throughs) than to the 70B aggregate. The cleaner read is that scale lifts detection F1 substantially while leaving replacement quality flat on automatic metrics, a dissociation we develop in the analysis below.

Single-call CoT detection F1 inverts at scale, on both phenomena. 8B SemEval idiom: detection-only sentence F1 is 0.333; the same model under dtr v2 jumps to 0.584 (+25pp from its detection-only baseline). 70B-AWQ SemEval idiom: detection-only sentence F1 is 0.529; the same model under dtr v2 *drops* to 0.496 (−3pp from its detection-only baseline). The same prompt that helps 8B by +25pp hurts 70B by −3pp on the same dataset. Mechanism is consistent with the VU metaphor inversion (Section 5, +10pp at 8B / −11pp at 70B): the 70B's token precision climbs when forced to commit alongside a paraphrase, but its recall drops as it over-commits to "no idiom" on borderline examples. The schema scaffold helps 8B and distorts 70B.

BLEU values are low overall (0.397–0.430), as expected: valid literal paraphrases of idiomatic expressions often share no n-grams with the gold reference. The idiom *smoking gun*, for example, may be correctly replaced with *decisive evidence* or *conclusive proof*, neither of which overlaps with the reference. BERTScore (0.914–0.916) is the more informative automatic signal here, capturing semantic proximity at the embedding level; the human-evaluation survey (Section 7.5) provides the definitive judgement on grammaticality, meaning preservation, and simplicity.

7.3 Metaphor Replacement (VU Amsterdam)

The VUAMC does not include gold literal paraphrases, making BLEU and BERTScore inapplicable. Replacement quality for metaphors is therefore assessed through (a) detection metrics on detect-then-replace runs (Table 5) and (b) qualitative inspection. Metaphor replacement is not in scope for the human-evaluation survey: as discussed in Section 2.5, the three-part rubric (grammaticality, meaning preservation, simplicity) is calibrated for idiom replacement, where the figurative-to-literal mapping is well-defined; conventional metaphors (*life is a journey*) often do not require replacement at all, and including them would dilute the rated pool.

Task	F1-sent	F1-tok	F1-span
Detection only	0.254	0.206	0.033
Detect-then-replace (v2)	**0.352**	**0.253**	**0.050**

Table 5: Metaphor detection metrics for detection-only vs. detect-then-replace (chain-of-thought, v2) runs on the full VUAMC evaluation set (16,202 sentences; Llama-3.1-8B-Instruct, $T = 0$). The detect-then-replace task lifts sentence F1 by 9.8 percentage points over detection-only. Runs: detection `2d322ad0`, detect-then-replace `491d332c`.

7.4 Analysis

BLEU is structurally misaligned with replacement evaluation. For tasks with multiple valid outputs, BLEU's n-gram overlap requirement penalises correct paraphrases that diverge lexically from the reference. BERTScore, which compares contextual embeddings, partially addresses this; but neither correlates reliably with human judgement on simplification-style tasks [2, 24], motivating the human-evaluation protocol below.

Idioms outperform metaphors at every replacement-relevant metric (H3). On the full evaluation sets the detect-then-replace sentence F1 is higher for idioms than for metaphors at both scales (idiom 0.584 at 8B and 0.496 at 70B; metaphor 0.352 at 8B and 0.190 at 70B), consistent with H3 and with the detection ordering reported in Section 5. The contrast is sharper still on the replacement side proper: on idioms the system produces a meaning-preserving literal paraphrase that can be scored against a gold reference (BLEU 0.40, BERTScore 0.91 on the gold-paraphrase subset) whereas the VUAMC has no gold paraphrase against which to score the metaphor rewrite at all: the absence of a measurable replacement metric for metaphors itself is consistent with H3's underlying claim that metaphor-to-literal mappings are less constrained. The human-evaluation survey covers idiom replacement only and will provide the definitive H3 evidence on the rubric dimensions.

The detect-then-replace prompt produces stronger detection than detection-alone on Llama-3.1-8B-Instruct. On both datasets, sentence F1 in detect-then-replace exceeds detection-only (VU metaphor: 0.254 $\rightarrow$ 0.352, +9.8pp; SemEval idiom: 0.333 $\rightarrow$ 0.584, +25pp). The structured-output schema in detect-then-replace appears to disambiguate the task for the model: forcing it to commit to a specific paraphrase pulls along sharper span identification. This is itself early evidence for RQ2 (decomposition > monolithic): even the trivial decomposition of *adding the replacement step* changes detection behaviour for the better at this scale. The same prompt change reverses sign on the 70B model (Section 5), so this is a small-scale effect, not a general one.

Scale (8B $\rightarrow$ 70B) lifts detection, with replacement quality deferred to the survey. Recomputing detection on the 27-source survey pool from the rated three-step-pipeline outputs, the 70B comparator detects a figurative expression in 24 of 27 sources against the 8B system's 21 of 27, and localises the span considerably more accurately (span F1 0.350 $\rightarrow$ 0.638 at IoU $\geq$ 0.5). Every survey-pool source is figurative, so sentence-level precision is fixed at 1.0 and sentence F1 reduces to detection recall (0.778 $\rightarrow$ 0.889); the detection comparison that includes negative examples is the full evaluation set (Table 4: sentence F1 0.333 → 0.529, +19.6pp). Reference-based replacement metrics are not reported on this pool. SemEval Task A supplies no literal-paraphrase reference, and the Task B paraphrase that exists for a minority of items is a distinct sentence sharing only the idiom, so BLEU and BERTScore against it measure cross-sentence overlap rather than replacement quality. The dissociation is therefore clean at this size class: scale sharpens the model's ability to *find* and delimit the figurative expression, while its ability to *rewrite* the expression is assessed by the human rubric (Section 7.6) rather than by automatic metrics.

7.5 Human Evaluation

The replacement task is evaluated by Direct Assessment [12, 13]; the protocol, rubric, standardisation, and three departures from the original DA specification are described in Chapter 4. This section gives the survey *instantiation*: conditions compared, source-sentence pool, item counts, and respondent recruitment. The full instrument (item layout, example items, and bad-reference construction recipe) is reproduced in Appendix 10.1, and the consent flow and data-protection statement in Appendix 10.3.

Conditions. The survey rates outputs from three open-weights conditions, all on the same set of source sentences:

- **Llama-3.1-8B + agentic 3-step pipeline** (detect → explain → transform), the deliverable system;
- **Llama-3.1-8B + monolithic single-prompt**, the RQ2 control condition, isolating the contribution of agentic decomposition;
- **Llama-3.3-70B-Instruct-AWQ + agentic 3-step pipeline**, the within-family scale comparator for RQ1.

No hosted-LLM condition is included in the human evaluation: the survey is reserved for the open-weights deliverable system and its controlled variants.

Pool and instrument. Sources are drawn from a candidate pool of 50 SemEval-idiom items selected from a Llama-3.1-8B detect-then-replace run, filtered for sentence length (< 30 words), expression length (≤ 4 words), and presence of a system-produced replacement; 30 sources were sampled by stratified random selection. Three sources were subsequently excluded: two during manual bad-reference review (the gold expression was judged literal in context) and one for contaminated gold annotation (the SemEval gold listed canonical idioms not present in the sentence). The final pool is therefore 27 sources, each rated under all three conditions: $27 \times 3 = 81$ items. Each respondent rates 20 items (16 rated + 4 quality-control), system-balanced and position-randomised per respondent; with a target of ≥ 3 ratings per item this requires at least 16 respondents. Respondents are native English speakers or non-native speakers with 10+ years of residence in an English-speaking country; recruitment statistics are reported with the results (Section 7.6).

Rubric. Three independent 0–100 continuous sliders per item: grammaticality, meaning preservation, simplicity (Section 2.5). Annotators are blind to the system-of-origin; system identifiers live in analysis-side metadata only.

Quality control. Two bad-reference items and two repeat-pair items per respondent, embedded among the 16 rated items (a 20% quality-control share), following the bad-reference and repeat-pair techniques of [12], which itself embedded a larger proportion of control items; fixed-threshold filters are used in place of [13]'s per-worker significance test, for the small-sample reasons documented in Section 4.2.3.

Aggregation. Per-annotator z-score standardisation before per-item aggregation, with mean ± 95% confidence interval reported for each rubric dimension and a composite summary across the three. Intra-annotator consistency is monitored through the repeat-pair absolute differences (Section 4.2.3).

7.6 Survey Results

The survey is the only instrument in this thesis that scores replacement quality directly, and it closes the measurement gap left by the automatic metrics above. The results are reported in four parts: the response and quality-control statistics (Table 7), the per-system rubric scores (Table 6), the within-family scale comparison that answers the RQ1 cut, and the free-text alternatives that respondents supplied.

	Raw means (0–100)			Standardised (z)			
System	**Gram.**	**Mean.**	**Simp.**	**Gram.**	**Mean.**	**Simp.**	**Comp.**
Llama-3.1-8B agentic	83.1	**81.6**	60.6	0.411	**0.336**	−0.378	0.123
Llama-3.1-8B monolithic	**85.4**	72.6	**64.1**	**0.520**	0.069	**−0.214**	0.125
Llama-3.3-70B agentic	78.4	76.4	51.9	0.253	0.162	−0.674	−0.086

Table 6: Human-evaluation results by system: raw 0–100 means and per-annotator-standardised (z) scores per rubric dimension, with the composite as the mean of the three standardised dimensions. 20 respondents after quality control; all 81 (source, system) pairs rated ≥ 2 times, 71 of 81 rated ≥ 3 times. The best value in each rubric dimension is shown in bold; the composite is not highlighted, as the two 8B conditions are statistically tied (Section 6.3).

Responses collected	22
native English speakers	19
long-term residents (10+ years)	3
Excluded: repeat-pair filter (mean $\lvert$diff$\rvert$ > 25)	2
Excluded: completion filter (< 10 of 20 items)	0
Flagged: bad-reference rule	0
Respondents retained	20
(source, system) pairs with ≥ 3 ratings	71 / 81
(source, system) pairs with = 2 ratings	10 / 81
Free-text alternative replacements collected	108

Table 7: Survey response and quality-control statistics (filters defined in Section 4.2.3).

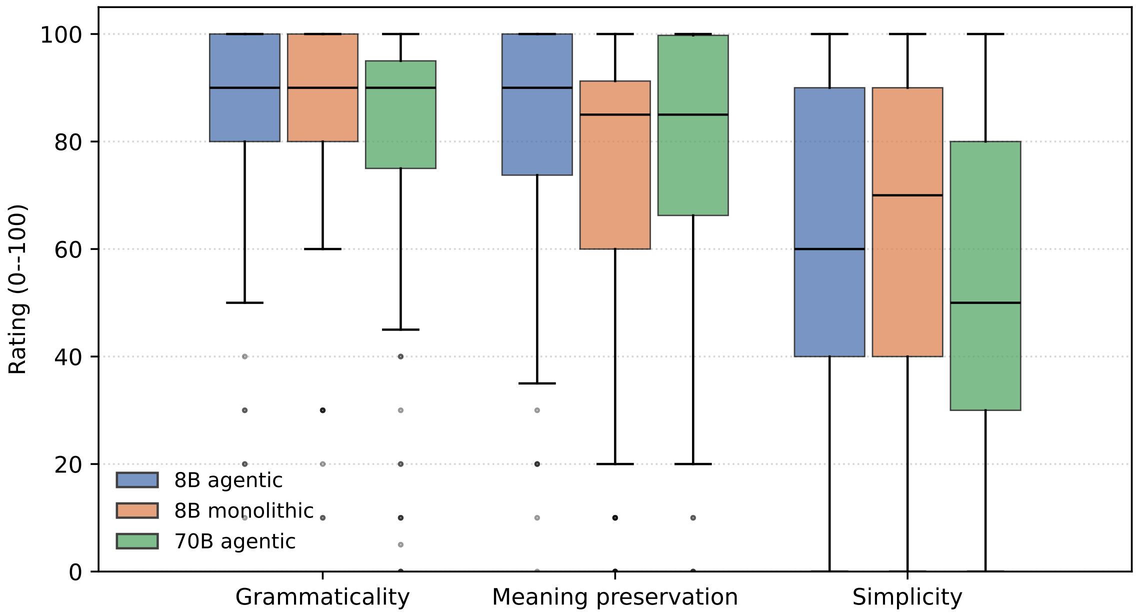

Figure 5: Distribution of raw 0–100 ratings per system and rubric dimension (20 retained respondents; bad-reference and repeat items excluded). Grammaticality and meaning preservation are high across all systems, while simplicity is the weakest dimension for every system and lowest for the 70B agentic comparator.

Response and quality control. Twenty-two forms were returned, 19 from native English speakers and 3 from non-native speakers resident in an English-speaking country for ten years or more (Table 7). Two respondents were excluded by the repeat-pair consistency filter, their mean absolute first-versus-second-showing difference exceeding 25 points on the 0–100 scale; none fell below the completion threshold, and none were flagged by the bad-reference rule. Twenty respondents were retained; the per-respondent quality-control outcomes are listed in Appendix 10.5. Their ratings cover all 81 (source, system) pairs at least twice and 71 of the 81 at least three times, meeting the target coverage on the large majority of items. The retained sample is small and predominantly native-speaker, so the analysis standardises ratings per annotator before aggregation (Section 4.2.3) to remove individual differences in scale use, and reports the system comparisons paired by source to control for item difficulty.

The systems preserve meaning but do not simplify. Across all three conditions, grammaticality and meaning preservation are rated highly while simplicity is rated substantially lower (Table 6; the full per-item breakdown is in Appendix 10.6). Raw grammaticality means fall between 78.4 and 85.4, and meaning preservation between 72.6 and 81.6, whereas simplicity ranges only from 51.9 to 64.1. The gap is consistent across systems: each loses roughly twenty to thirty points moving from the grammar and meaning dimensions to simplicity. After standardisation, simplicity is the one dimension on which all three systems score below the annotator mean (z between −0.214 and −0.674), while grammar and meaning remain above it throughout.

The systems produce outputs that are fluent and that retain the meaning of the source sentence, but that are not perceived as easier to read. For Easy-to-Read adaptation this is the dimension that matters most, and it is the one the systems leave unaddressed. The finding also exposes the limit of the automatic metrics reported above: BERTScore sat near 0.915 for every system and could not separate them, yet the human rubric distinguishes a meaning score of 81.6 from a simplicity score of 60.6 within a single system. The quantity the deliverable most needs to improve is precisely the quantity that no reference-based metric in this study could see, which is the central reason a human evaluation was required rather than optional.

On the composite the two 8B conditions are indistinguishable (z 0.123 versus 0.125). The per-dimension differences between the agentic and monolithic conditions, which bear on RQ2, are taken up in Section 6; the present section reads the three systems descriptively and turns to the scale comparison.

Scale does not buy human-perceived quality. The within-family scale comparison sets the 70B agentic pipeline against the 8B agentic pipeline on the same sources, paired by source ($n = 27$). The 70B comparator is rated directionally lower on all three raw dimensions (grammaticality 78.4 versus 83.1, meaning 76.4 versus 81.6, simplicity 51.9 versus 60.6) and lower on the composite (z -0.086 versus 0.123). The difference reaches conventional significance on one dimension, simplicity, where the 70B system scores 0.347 standardised units below the 8B system (Wilcoxon $p = 0.026$, paired t $p = 0.075$); the composite difference is marginal (Wilcoxon $p = 0.066$). Read together with the automatic-metric result above, where scale lifted detection but not replacement, the rubric ratings give a consistent verdict: in this setup, scale did not buy, and on simplicity may have cost, human-perceived quality. The claim is deliberately narrow. It rests on one dimension, one model pair, and twenty raters, and it concerns perceived simplicity rather than replacement quality in general. What it establishes is that the larger model, which detects and delimits idioms more accurately, produced rewrites that annotators found no easier, and on simplicity somewhat harder, to read. Scale improved the part of the task that automatic metrics already captured and did not improve the part that only human judgement could reach.

Annotators supplied their own replacements. Respondents could add a free-text comment to any item, and did so on 93 rated items. Most of these comments offered the respondent's own replacement rather than only flagging a problem, and those alternatives were consistently shorter than the system outputs. Section 7.7 examines them in detail; the pattern there, that human readers compress and plain-word the figurative expression where the systems expand it into a gloss, is the qualitative face of the simplicity gap reported above.

7.7 Free-Text Responses from Annotators

Each survey item carried an optional free-text box. Annotators used it on 93 rated items, and most of those comments did more than register a complaint: 56 of the 93 proposed a replacement of their own, against 37 that only flagged a problem. The proposed replacements are the more useful half, because they show what a human reader does with the same sentence the system was given. A larger sample of these alternatives is reproduced in Appendix 10.7.

One pattern is consistent across these suggestions: the human replacement is shorter than the system's, and reads more naturally. For the idiom "out of control", the 70B system produced "the radio crime wave got so *became too chaotic or unmanageable to handle* that", which both over-explains the phrase and breaks the grammar of the sentence; the annotator simply wrote "became too chaotic". For "all-rounders", the 8B system wrote "business people with a wide range of skills and responsibilities" where the annotator offered "generally skilled in the business field". For "be the role model", the monolithic system produced "a person she looked up to as a model, but she never had one like that" against the annotator's "looked up to".

The reason is the same in each case. The systems treat replacement as a substitution at the site of the figurative expression, and they tend to paste in a dictionary-style explanation of its meaning. This keeps the meaning but lengthens the sentence, and now and then, as in the "out of control" case, leaves it ungrammatical. A sharper instance is the 70B rendering of "copy cat" in "the SB silhouette brought *copy cat* design language": the system produced "the

SB silhouette brought *imitate or copy the style of someone or something else* design language", dropping a verb phrase where a modifier was needed, while the annotator rewrote it as "matches the style of". Human readers do not substitute in place; they rewrite the phrase into a short, plain equivalent that fits the sentence.

The 37 comments that only flag a problem are informative too. Some point to sentences the system left unchanged, where its detection step missed the idiom (Section 6.4); others object to the length of the system's explanation. Both complaints point the same way.

Taken together, the free-text suggestions name the gap the rubric measures. The systems produce output that keeps the meaning; human readers, given the same sentences, compress and plain-word it. That difference, expansion against compression, is the qualitative face of the simplicity shortfall in Section 7.6, and it shows where the next gain in Easy-to-Read quality lies: not in finding or preserving meaning, which the systems already do, but in producing a shorter, plainer rewrite.

7.8 Qualitative Examples

The aggregate scores and the free-text patterns above are clearest on individual items. Three replacements from the survey pool, all produced by the 8B deliverable, span the range the rubric measures: a clean success, the system's typical half-success, and an outright failure. The scores quoted are the per-item raw means across respondents, on the 0–100 scale.

A clean replacement. For the source "Many of those who escaped the genocide like her have borne numerous sufferings since, adding to the heavy cross they already have to bear", the pipeline detected *bear the cross* and produced "Many of those who escaped the genocide like her have endured or coped with numerous sufferings since, adding to the heavy situation they already have to cope with". The idiom is gone, the meaning is preserved, and the sentence still reads naturally, even though the idiom appeared in a split form ("the heavy cross they already have to bear") that the system had to recognise and rewrite as a unit. Respondents rated it 91 for grammaticality, 88 for meaning preservation, and 95 for simplicity. This is the behaviour the rubric rewards, and it is what the systems achieve when the figurative span is a familiar, self-contained idiom.

Meaning kept, but no simpler. The more common outcome preserves meaning without making the sentence easier to read. For "Fox News is no longer the top dog", the pipeline detected *top dog* and produced "Fox News is no longer the most powerful or dominant entity". Nothing is wrong with the rewrite: it is grammatical and faithful, and it scored 98 and 97 on those two dimensions. But it trades a short, vivid two-word idiom for a five-word formal description, so its simplicity fell to 47. This is the expansion pattern of Section 7.7 in miniature, and it is the single behaviour most responsible for the low simplicity scores across the survey.

An over-explained replacement. When the idiom's meaning is hard to state briefly, the substituted gloss can run long enough to damage the sentence. For "Trudeau dismissed decriminalization last year, telling the Canadian Broadcasting Corp it was not a silver bullet", the pipeline replaced *a silver bullet* with "a solution that is expected to solve a problem easily, but may not be effective in reality". The clause is too long for its slot, and its hedged second half clouds the meaning the idiom carried, so the item scored only 51 for grammaticality and 60 for meaning, with simplicity at 24. Failures of this kind are not detection errors (the idiom was found correctly) but transformation errors: the system knows what the phrase means and cannot say it briefly.

Across the pool these three outcomes recur in this order of frequency: faithful clean rewrites and faithful-but-long rewrites are common, while outright damage is rarer. The distribution in Figure 5 is the aggregate of exactly this mix, and it places the remaining work on the transformation step: learning to say briefly what the system already understands.

8 Discussion & Limitations

8.1 Synthesis: Structure over Scale

Finding	At 8B	At 70B	What it shows
(i) Adding a forced replacement field to the detection prompt	+10 / +25pp sentence F1 (VU / SemEval)	−11 / −3pp	the same prompt change helps the small model and hurts the large one
(ii) Pipeline detection vs. detection alone	identical (3 decimals)	identical (3 decimals)	splitting the task into steps does not change detection
(iii) Human-rated replacement quality	8B agentic composite $z = 0.123$	70B agentic $z = -0.086$; simplicity $p = 0.026$	a larger model is rated no better, and worse on simplicity

Table 8: The three findings. Each is an effect that common practice would expect to hold across models, and does not. Together they motivate the central claim that explicit task structure is the reliable lever in this task, and that human evaluation is what makes its effect visible.

This thesis asked where reliable quality comes from when large language models rewrite figurative language into Easy-to-Read English. Common practice expects two things to drive the gains: a larger model and a better prompt. Table 8 collects three results that go against this. Each is an effect that practice would expect to hold across models, and none of them does.

The first concerns the prompt. Adding a forced replacement field to the detection prompt raises detection on the 8B model, by ten to twenty-five points, but lowers it on the 70B model, by three to eleven points, with the same prompt and the same data (finding i). The same change helps the small model and hurts the large one. The second concerns task structure read narrowly: the three-step pipeline detects as well as a plain detection prompt, matching it to within rounding on every model and dataset (finding ii). Splitting the task into steps does not make detection sharper. The third concerns scale: the 70B model, which finds and marks idioms more accurately, is rated no better overall by human readers and worse on simplicity (Wilcoxon $p = 0.026$). In this setup, scale did not buy, and on simplicity may have cost, human-perceived quality (finding iii).

These three results are connected. They show that model size and prompt wording are unreliable levers in this task: their effects do not just vary in size, they can reverse from one model to the next, and the automatic metrics that might catch the reversal cannot see replacement quality at all (Section 6). Someone who tunes a prompt on one model, or who assumes a larger model is a better one, is trusting a signal that can flip on the system they actually ship.

Explicit task structure behaves differently, and that is the claim of this thesis: structure is the lever that can be relied on, not the lever that always wins. Two things follow from breaking the task into visible steps. First, the decomposition is what made the results above measurable at all. The reversal in finding (i) and the equal detection scores in finding (ii) are statements about one specific step, and only a system that exposes its steps can report them. Second, where structure helps, the help can be traced: its benefit shows up on the replacement side, as a consistent directional advantage on meaning preservation (+0.31 standardised units, p =

0.06-0.08) that approaches but does not reach conventional significance (Section 6.3), and every failure can be tied to a named step rather than to one opaque rewrite (Section 8.4). Human evaluation is not itself a lever on quality; it is how that quality is made visible, because no automatic metric in this study could see it.

The limits of the claim are worth stating plainly. Decomposition helps where the task is open-ended, which is rewriting a figurative expression into a literal one, and not where it is a yes-or-no decision, which is detection: finding (ii) shows this directly, and the overall human rating of the agentic and monolithic systems is tied (z 0.123 vs. 0.125), so structure is not a free win on every measure. What the three findings support together is narrower, and more useful for a real system, than a leaderboard score. In this task the gains people expect from scale and from prompt wording are model-dependent and can reverse, while explicit task structure is the lever whose effects are visible, traceable, and aimed at the dimension Easy-to-Read adaptation can least afford to lose. That is what structure over scale means here: not that structure always wins, but that it is the lever that can be trusted to mean what it says.

8.2 The Simplicity Gap

The headline survey result, that every system scores high on meaning preservation but low on simplicity (Section 7.6), has a consistent mechanism behind it, visible in the free-text alternatives respondents supplied (Section 7.7). When the systems literalise a figurative expression, they tend to substitute a *definition* of it: a clause that spells the meaning out, faithful but longer than the original. “All-rounders” becomes “business people with a wide range of skills and responsibilities”; the 70B renders “poison pill” as “a financial strategy to prevent a hostile takeover by making the company less attractive to potential buyers”. The meaning is carried over, which the rubric rewards, but the sentence has grown and its register has not become plainer, which the simplicity axis penalises. Human readers, given the same sentences, do the opposite: they *compress*, reaching for a short, plain near-synonym (a respondent rewrote “all-rounders” simply as “generally skilled in the business field”) rather than a gloss.

The gap the survey measures is therefore expansion against compression. It is not that the systems fail to understand the expression, their meaning scores show that they do, but that literalisation-by-definition is the wrong operation for Easy-to-Read, whose goal is brevity and plainness rather than a fuller restatement. This is why simplicity, and not meaning, is the dimension on which the deliverable falls short, and it is part of the reason a larger model does not help: a 70B writes longer, more careful definitions, which raise faithfulness while leaving, or worsening, the readability the task is about. The clearest direction for future work follows directly (Section 8.7): a transformation step optimised for plainness, scored against a simplicity objective, rather than one optimised only to preserve meaning.

8.3 Why Metaphors Resist Literal Replacement

The synthesis above located structure’s benefit on the replacement side of the task. That benefit is in turn bounded by the linguistic material itself, which is where this section turns. Hypothesis H3 held that idioms are more tractable than metaphors for figurative-to-literal replacement, and the evidence of Chapter 7 supports this indirectly: the idiom track can be scored against literal-paraphrase references at all, whereas the metaphor track has no measurable replacement metric, and that absence is itself the symptom. The deeper reason is linguistic rather than architectural. An idiom is a conventionalised expression whose figurative meaning is fixed by usage: “spill the beans” means reveal a secret, and that mapping is stable across contexts. Because the target meaning is determined, a literal paraphrase exists and can be written down, which is why the SemEval idiom data can supply reference paraphrases for a subset of items at all.

Metaphor is different in kind. A conceptual or novel metaphor maps a source domain onto a target in a way that can carry information no single literal sentence preserves; the figurative phrasing does work that paraphrase discards. This shows up in the data as an absence. The VU Amsterdam Metaphor Corpus (VUAMC) [25] annotates metaphor at the token level, following the MIPVU procedure [26], but supplies no gold literal paraphrase, and that gap is not an oversight: there is often no single literal restatement to annotate, because the mapping from a metaphor to what it "literally means" is underdetermined.

The dependence runs along conventionality. Neidlein et al. [22] show that pre-LLM metaphor-recognition systems earn most of their headline F1 on conventionalised metaphors and frequent word types, and degrade sharply on unconventional metaphors and rare words, so that much of the apparent competence is conventionalised word-sense disambiguation rather than metaphor modelling. Read for the replacement task, this stratifies the problem. A conventionalised metaphor behaves like ordinary vocabulary: its figurative sense is already part of everyday language, so it tends to be detected well and often needs no rewrite for a reader who already knows the sense. A novel metaphor is where both steps degrade at once: it is harder to detect, and there may be no literal target to rewrite it into.

Two consequences follow for this thesis. First, the human-evaluation survey covers idioms only (Chapter 4). This followed from the data rather than from a free choice: the VUAMC provides no gold paraphrases to evaluate against, and the meaning-preservation rating only makes sense when there is a recoverable meaning to preserve, which idioms have and many metaphors do not. Second, the metaphor track is therefore evaluated through detection metrics alone, and a faithful metaphor-replacement evaluation would need a different instrument, one that does not assume a single gold paraphrase and that can score a rewrite against a space of acceptable readings. That instrument is left to future work.

8.4 Observability and Error Analysis

Observability began as a fourth research question and is treated here as a qualitative discussion rather than a measured result: no formal observability metric was defined or evaluated. The argument is nonetheless tied to the central one, because the inspectability that decomposition provides is what made the preceding results possible to state at all.

The agentic pipeline records its intermediate state. Each run stores the detected expression, the explanation of its meaning, and the produced replacement as separate fields, so a failure localises to a step: a missed expression is a detection error, while a correct detection followed by a meaning-distorting rewrite is a transformation error, and the two are distinguishable in the trace. The monolithic condition offers no such resolution. Its output is a single free-text string, so an error is only ever "the rewrite is wrong", with no internal state to say why.

A concrete case makes the difference visible. On the source "Cubic announced a poison pill, for one year, allowing investors taking stakes above this threshold to be diluted", the 8B pipeline returned the sentence unchanged. The trace shows why: the detection step produced no span, so the transform step short-circuited and passed the input through untouched. The failure is attributable to a single component, the detection prompt, and a fix can target it without disturbing the transform step that was never reached. The unchanged output also exposes a measurement trap. A sentence returned verbatim is, by construction, fluent and faithful to the source, so it scores well on grammaticality and meaning preservation while having performed none of the task; only the simplicity dimension registers the non-event. The same outcome from the monolithic system would be opaque: when a single-prompt model returns its input unchanged, there is no way to tell whether it failed to recognise the idiom or judged that no change was needed. On this particular item the monolithic system happened to rewrite "poison pill" as "defensive measure" and the 70B pipeline detected and rewrote it, so decomposition bought no advantage in output here; what it bought was a diagnosis. Only the pipeline conditions can report, for any given item, which step made the decision. Figure 6 shows the three behaviours side by side.

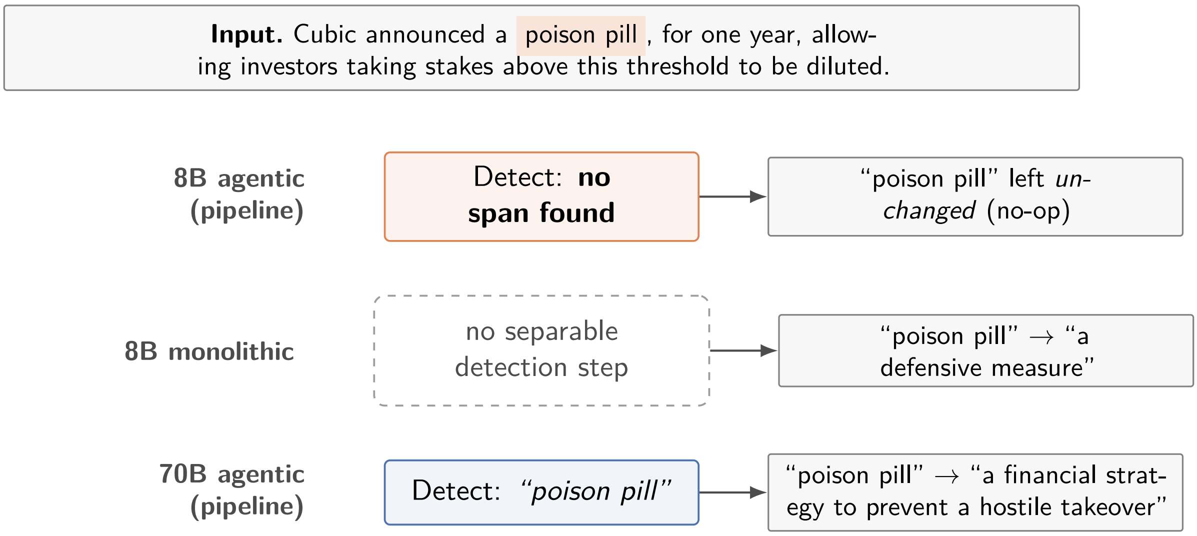

Figure 6: The same input under the three systems. The pipeline conditions (8B and 70B agentic) expose a detection step whose result is inspectable: the 8B pipeline records that no span was found and so returns the sentence unchanged (a diagnosable no-op), while the 70B pipeline detects the idiom and rewrites it. The monolithic system has no separable detection step (dashed), so although it happens to rewrite the expression, there is no internal state to say whether it recognised the idiom or merely chose to reword. Structure does not guarantee a better output here, but it makes the behaviour attributable to a step.

This is the methodological payoff of structure, and it underwrites the synthesis of Section 8.1. Two of the three findings, the prompt-change reversal (finding i) and the equal detection scores (finding ii, Section 6.4), are statements about the detection step in particular, which the decomposition makes a separate part that can be measured on its own; a single-output system gives no such step to measure or report against. Observability is therefore not a benefit running alongside the quality results, but the property that let those results be pinned to a component at all.

8.5 Limitations

Conceptual limits. Two conceptual limits bound the task: some metaphors encode meaning that cannot be fully literalised, and replacement can oversimplify nuance. The choice to scope the human-evaluation survey to idioms only (Section 2.5) reflects this limit pragmatically: the three-part rubric of grammaticality, meaning preservation, and simplicity is well-defined for the figurative-to-literal mapping in the SemEval idiom set, but breaks down for conventional metaphors that may not require any rewrite at all.

Evaluation limits. The subjectivity of human evaluation, mitigated by per-annotator z-score standardisation [12] and the assessor-intrinsic QC techniques of bad-references and repeat-pairs [13], remains an inherent property of the task. There is no single ground-truth literal paraphrase for any given idiom, and BLEU's structural misalignment with the task (Section 7) means the survey's ratings across the three rubric dimensions (grammaticality, meaning preservation, simplicity) are the load-bearing evidence for replacement quality. The respondent pool (22 collected, 20 retained after quality control), while sized for ≥ 3 ratings on 71 of the 81 items, is at the lower end of what Graham et al. [12] found stable for per-annotator standardisation; this is documented as a deliberate departure in Section 2.5. Three of the twenty retained respondents are long-term residents of English-speaking countries rather than native speakers.

Model dependence. *Findings observed under one model do not necessarily transfer to another, and may invert.* Two instances of this appear in the thesis, in increasing order of severity:

(1) The detect-then-replace detection lift inverts with scale, on both phenomena. On Llama-3.1-8B-Instruct, the dtr prompt's structured-output schema gives a +10pp sentence-F1 lift over the model's own detection-only baseline on VU metaphor (0.254 → 0.352) and a +25pp lift on SemEval idiom (0.333 → 0.584): the auxiliary-replacement field forces the model to commit detected spans alongside a paraphrase, scaffolding detection at small scale. On Llama-3.3-70B-Instruct-AWQ, the same prompt gives an 11pp sentence-F1 *loss* from detection-only on VU metaphor (0.302 → 0.190) and a 3pp loss on SemEval idiom (0.529 → 0.496). Both datasets, same direction, same prompt, no other change in conditions: +10 to +25pp at 8B, −3 to −11pp at 70B. The mechanism is consistent across phenomena: the schema constraint scaffolds the under-specified 8B, but on the better-calibrated 70B it forces over-commitment to "no figurative content" on borderline items, so the larger model becomes more selective (its token precision climbs) at the cost of recall and ends up worse overall.

(2) Pipeline detection is architecture-invariant. As reported in Section 6.4, the 3-step pipeline's detection step scores within rounding of the standalone detection prompt across models and datasets (the small residual differences trace to differing structured-output failure sets, not to the detection step itself). This pins the detection lift, and its inversion in (1), to the structured-output schema's commitment mechanism rather than to chain-of-thought content, and implies that whatever the pipeline architecture buys is in the replacement step's quality, not in detection sharpness.

These two findings cohere into a single methodological claim: *a prompt or design choice validated on one model can fail, or reverse, on another, so a result obtained on a small open-weights model gives an unreliable read on what helps the same family at a larger scale.* The risk of taking such a result at face value is shipping a system whose design choices were tuned to one model's failure modes and do not help, or actively hurt, at the scale actually deployed. Tian et al. [30] (TSI), Gao et al. [9] (PASS), and Badran et al. [4] (the sequential SemEval Task 1A pipeline) all report architectural wins on larger backbones; the thesis's pipeline comparison is a partial re-validation on an open-weights deliverable, and finds the wins preserved on the detection-replacement split but *not* as detection improvements at scale.

Scale dependence (within family). The within-family comparison (Llama-3.1-8B vs. Llama-3.3-70B-Instruct-AWQ) shows that scale helps detection unevenly and does not help replacement at all. On the full evaluation sets, the larger model lifts idiom detection substantially (sentence F1 0.333 → 0.529, +19.6pp) but metaphor detection only marginally (0.254 → 0.302, +4.8pp; Section 5). The asymmetry has a clear cause. Metaphor detection is scored against MIPVU's exhaustive token-level gold, and the 70B's gain there is almost entirely in precision (token precision 0.238 → 0.355) with recall nearly flat (0.183 → 0.212), so the larger model is more selective about the metaphors it flags without marking many more of the conventional ones the gold counts; idiom detection, which targets a single discrete expression, is where scale genuinely pays. The 27-source survey pool shows the same direction for idioms, recomputed from the rated pipeline outputs: the 70B detects a figurative expression in 24 of 27 sources against the 8B's 21 and localises spans more accurately (span F1 0.350 to 0.638). Replacement quality, by contrast, does not improve with scale: reference-based metrics are not well defined for this task (Section 6), and the human evaluation rates the 70B directionally lower on every rubric dimension and significantly lower on simplicity (Wilcoxon $p = 0.026$; Section 7.6). In this setup, scale did not buy, and on simplicity may have cost, human-perceived quality. The 70B comparator is therefore most defensible as a detection-side ceiling for idioms rather than a replacement-side competitor. One caveat bounds even that reading: the 70B comparator (Llama-3.3) is a later model generation than the 8B (Llama-3.1), so this comparison conflates scale with a generation improvement and its detection gap is an upper bound on the effect of size alone.

Scope limits. The work stays at the sentence and multi-word-expression level throughout, with no discourse-wide figurative-language tracking, and is primarily in English (the FACILE umbrella's Spanish work [27] is the natural extension target). The replacement evaluation covers idioms only; metaphor replacement is assessed through detection-side metrics on detect-then-replace runs, not through human-rated rubric scoring.

8.6 Hypothesis H4: Untested

Beyond the bounded claims above, one hypothesis was set aside entirely. Hypothesis H4 held that *explicit intermediate explanations generated by the agent correlate positively with the semantic correctness of the final literal replacement*. It was not tested in this thesis, and the reason is stated plainly rather than left implicit. Testing the hypothesis means operationalising it as a correlation between the *quality* of each explanation and the correctness of the replacement it produced, which requires a score for explanation quality, and no such instrument was built: the survey rates the final replacement on grammaticality, meaning preservation, and simplicity, but never the explanation that produced it. Reporting a correlation without an explanation-quality measure would be a claim without evidence.

The raw material for the test nonetheless exists. Every agentic run persists the explanation alongside the detected expression and the replacement as structured output (Section 8.4), and each rated agentic item carries a human meaning-preservation score. What is missing is only the explanation-side score. A correlation study is therefore the natural follow-up: rate explanation quality independently, either through a second Direct Assessment pass or through an LLM-as-judge scheme with human spot-validation, then correlate the per-item explanation scores against the meaning-preservation ratings. The unit of analysis is the source within an agentic condition, of which the 8B deliverable has 27, so the design should begin with a power analysis at $n = 27$: the pool is small enough that a weak correlation might not be detectable, which is precisely why the test was not bolted on post hoc to the present survey. Stating H4 as open, with a concrete design attached, is the honest position given the evidence actually collected.

8.7 Future Work

- **Mixture-of-experts evaluation**: Llama 4 (Scout, Maverick) was excluded from this work because the smallest variant exceeds the 40 GB A100's VRAM. With access to A100 80 GB or H100, the within-family scale comparison could be extended into a within-architecture-class comparison, testing whether the agentic decomposition's value transfers to MoE backbones that have different reasoning-vs-throughput tradeoffs.

- **Decomposition variants**: the no-explanation and no-self-verification variants (Section 6) are deferred from this thesis to keep the human-evaluation survey one-shot; isolating the contribution of each agentic step is the natural next study.
- **H4 correlation study**: the explanation-quality versus replacement-correctness design sketched in Section 8.6, over the persisted structured outputs and the existing survey ratings.
- **A human-evaluation instrument for metaphor replacement**: the idiom rubric does not transfer (Section 2.5); a metaphor instrument would need conventionality stratification and a "no rewrite needed" option before rubric rating.
- **Curriculum-style data augmentation**: [18] reaches SOTA on metaphor detection with single-step inference at test time after curriculum-augmented fine-tuning. If the open-weights detection ceiling at 8B (sentence F1 0.254 on VU metaphor detection) is judged insufficient for downstream E2R deployment, CDA-style fine-tuning is the next intervention to try before declaring the open-weights tier insufficient.
- **Spanish FACILE integration**: the per-guideline-module pattern of the UPM E2R cluster [27, 28] is language-agnostic at the architectural level. Porting the figurative-language replacement module to Spanish is contingent on a Spanish gold-paraphrase dataset; such a dataset does not currently exist for idioms, so the immediate work is dataset creation rather than model adaptation.

9 Conclusion & Future Work

9.1 Answers to the Research Questions

This thesis asked whether an agentic large-language-model pipeline can detect figurative language and rewrite it into easier-to-read English, and where the quality of that rewriting comes from. The three research questions are answered as follows.

The first question asked how well a zero-shot agentic pipeline detects and interprets idioms and metaphors, compared with supervised systems. The answer is partial. The pipeline detects figurative language but stays below systems trained for the task. The comparison here is indicative rather than a leaderboard ranking, since the zero-shot setting, the metrics, and the test splits all differ from the supervised work, and supervised systems can still win on these benchmarks [18]. Within the open-weights family, a larger model helps detection, but unevenly across phenomena: idiom sentence F1 rises by almost twenty points from 8B to 70B (SemEval 0.333 → 0.529), while metaphor sentence F1 rises by under five (VU 0.254 → 0.302), because metaphor is scored against an exhaustive token-level gold that the larger model recovers little more of. The gain from prompt wording is less stable still. Forcing the detection prompt to also commit to a replacement raises 8B detection by ten to twenty-five points but lowers 70B detection by three to eleven, the same change with opposite signs. Scale lifts detection, but its gains end there (Section 5).

The second question asked whether breaking the task into explicit steps, detection then explanation then transformation, improves on a single prompt. The answer is suggestive support, not a confirmed effect. No automatic metric could settle the question, since BLEU and BERTScore disagree in direction on the same outputs. The human evaluation gives the agentic pipeline a consistent directional advantage on meaning preservation (+0.31 standardised units, $p = 0.06$-0.08) that approaches but does not reach conventional significance, while the overall composite score is tied with the monolithic baseline (z 0.123 vs. 0.125). The clearer dividend of decomposition is not a higher score but traceability: because the pipeline exposes its steps, every failure can be tied to a named step rather than to one opaque rewrite (Section 8.4). Decomposition helps where the task is open-ended, which is the rewriting, and not where it is a yes-or-no decision, which is the detection (Section 6).

The third question asked how well the system replaces figurative expressions with literal, meaning-preserving alternatives, and what trade-off arises between fidelity and readability. The systems preserve meaning but do not simplify. Human raters scored grammaticality and meaning preservation high across all systems (raw means in the seventies and eighties) but simplicity low everywhere (in the fifties and low sixties), the weakest dimension in every condition. Idioms were replaced more reliably than metaphors, as expected (H3); metaphors have no gold paraphrases to evaluate against, which is itself part of the finding rather than a gap in the data. Scale did not rescue the weak dimension: the 70B model was rated significantly worse on simplicity (Wilcoxon $p = 0.026$), so in this setup, scale did not buy, and on simplicity may have cost, human-perceived quality (Section 7).

The fourth hypothesis, that the agent's intermediate explanations correlate with the correctness of its replacements (H4), is stated but not tested in this thesis. The study needed to measure it, and the design for that study, are set out as future work in Section 8.6.

9.2 Contributions and Outlook

The thesis makes four contributions. It delivers an agentic open-weights pipeline for figurative-language Easy-to-Read adaptation, with a monolithic single-prompt system as a designed control, with all experiments persisted and reproducible. It adapts the Direct Assessment human-evaluation protocol to this task, with its departures from prior practice documented, because no automatic metric scores figurative-language replacement reliably. It reports what does and does not transfer in this task: scale lifts detection but not human-perceived replacement quality, prompt gains can reverse in sign across model sizes, and decomposition buys a consistent directional advantage on meaning preservation. Finally, it leaves behind a dataset of 108 human-written alternative replacements and rubric ratings for 81 system outputs, for future work.

The findings carry a practical lesson for anyone building such a system. The common reflexes, reach for a larger model or tune the prompt, are exactly the levers this study found unreliable. The prompt change that helped the 8B model hurt the 70B one, so a prompt validated on a small development model can give a misleading read on what will help the model that is actually shipped. A larger model detected figurative language better yet was rated no better, and worse on simplicity, by human readers. The lever that behaved predictably was explicit structure: breaking the task into named steps did not raise the score on its own, but it made the system's behaviour measurable and every failure attributable to a step. For a system meant to serve readers who depend on accuracy, that inspectability is worth as much as a metric gain, because it is what lets a human catch an error before it reaches the reader.

The wider implication is for accessible-text research. Easy-to-Read guidelines say to remove figurative language but not how, and this thesis supplies the missing piece as one module in a per-guideline pattern of automated adaptation, alongside the modules for morphological features and dialogue. It also marks out the work still to be done. The clearest weakness was simplicity: the systems kept meaning but did not make the text easier to read, which is the one thing an Easy-to-Read tool exists to do. Closing that gap is the most important next step, and it points to a rewriting stage trained or prompted for plainness directly, with an evaluation that weights simplicity rather than treating it as one dimension among three. Metaphors mark out a second piece of unfinished work. They have no gold paraphrases to score against, so progress on them is blocked less by the model than by the lack of an evaluation instrument; building a reference set and a rubric for metaphor replacement would let the field measure what it currently can only inspect by hand. A third step returns to the original motivation: porting the pipeline from English to Spanish, where the FACILE programme already operates, would test whether the structure-over-scale result holds across languages and move the work toward the readers it was built for. The explanation-quality question (H4) remains open and can be settled by the correlation study set out in the Discussion.

Across two figurative phenomena, two model scales, an agentic pipeline set against a monolithic baseline, and a twenty-rater human evaluation, the reliable lever for making figurative language easy to read was never model size or prompt cleverness: it was explicit structure, in the system that produced the text and in the evaluation that judged it.

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10 Annex

10.1 Survey Design (Direct Assessment, Idiom Replacement)

This appendix reproduces the full survey instrument described in Chapter 4 (protocol) and Section 7.5 (instantiation). It documents the per-item layout, per-respondent flow, quality-control item construction, fixed-threshold filter values, and the form-deployment choice.

10.1.1 Per-item layout

Each rated item presents the original sentence, the system-detected idiomatic expression, and the system's full-sentence replacement, followed by three independent 0–100 sliders. An optional free-text field captures alternative paraphrases for qualitative analysis.

```
Original sentence:
  "Unni, the stylist, is on cloud nine after having an opportunity to
   style the beard of his favourite star."

Idiomatic expression detected by the system:
  "on cloud nine"

System's replacement of the full sentence:
  "Unni, the stylist, is extremely happy after having an opportunity to
   style the beard of his favourite star."

Please rate the system's replacement on three dimensions
(0 = unacceptable, 100 = perfect):

  Grammaticality (is it correct English?)             [ 0 -- 100 ]
  Meaning preservation (same meaning as original?)    [ 0 -- 100 ]
  Simplicity (clearer than the idiom version?)        [ 0 -- 100 ]

Optional, if the replacement misses the mark,
what would you write instead?

  [                                                                  ]
```

The detected expression is shown explicitly so annotators do not have to find the idiom themselves; this controls for "did the annotator notice the idiom?" as a confound. The SemEval Task B reference paraphrase is *not* shown: Task B references are comparison sentences, not meaning-equivalent paraphrases, and showing them would distort the meaning-preservation rating.

10.1.2 Example rated items

The three system conditions for one source sentence are shown below, drawn from the live survey, to illustrate the comparative design. Respondents saw each replacement on its own screen, in randomised order and without system labels (the survey was blind); the labels here are for documentation only.

Source. "Most accountants are business *all-rounders* for whom number crunching tax issues occupy but a small part of their day."
Detected idiom. *all-rounders*

8B agentic (pipeline). Most accountants are business people with a wide range of skills and responsibilities for whom number crunching tax issues occupy but a small part of their day.

8B monolithic. Most accountants are people who work in business and have many different skills, and calculating tax problems is only a small part of what they do every day.

70B agentic (pipeline). Most accountants are business people with a wide range of skills and expertise for whom number crunching tax issues occupy but a small part of their day.

Each replacement was rated on grammaticality, meaning preservation, and simplicity using the layout above. The example also shows a pattern the survey results bear out: the rewrites preserve meaning but tend to retain the source's complexity, and the larger model is not visibly simpler.

10.1.3 Per-respondent flow

1. **Intro page**: plain-language task description, ethics paragraph (supplied by Mari Carmen Suárez-Figueroa), and a native-English-speaker self-attestation. No name or email is collected; the only demographic field is the binary native-English-speaker check.

2. **Calibration item**: one worked example before the rated items, with explanations of each rubric dimension. Not graded; for orientation only.

3. **Rated items (16)**: random subset of the 81-item pool (27 sources × 3 conditions). Sampling is constrained so each respondent sees a mix of the three system conditions and never the same source twice except as a deliberate repeat-pair (Section 10.1.4).

4. **QC items (4)**: two bad-reference items and two repeat-pair items, interleaved among the 16 rated items at randomised positions. This gives 20 items per respondent, of which the four QC items are 20%.

5. **Submission**: on submit, raw scores are written to a Google Sheet for offline analysis. No real-time feedback is shown to the respondent.

Estimated completion time is 22–25 minutes. A two-respondent pilot is run before live distribution to verify median completion time; if median exceeds 25 minutes the rated-item count is cut to 12 (with the QC count held at 4) before going live.

10.1.4 Quality-control item construction

Bad-reference items. Two of each respondent's 20 items are constructed by deliberate degradation of a system output, anchored on *meaning preservation* as the most binary and least ambiguous of the three rubric dimensions. The degradation strategies, in order of preference:

1. Replace the whole sentence with a sentence whose meaning differs from the original (default).

2. Replace the idiom with a meaningless or syntactically broken phrase (breaks grammaticality instead of meaning).

3. Replace the idiom with a more obscure idiom (breaks simplicity instead of meaning).

Bad references are generated by a separate LLM run with a “make this output worse on dimension X” prompt and manually reviewed by the author before survey distribution. Each respondent sees one bad reference of strategy 1 and one of either strategy 2 or 3, balanced across the respondent pool.

Repeat-pair items. Two items in each respondent’s set are duplicates of two other items in the same set, with at least five other items separating each pair’s two showings. The pairs are randomised per-respondent so the same source items are not always re-rated, avoiding bias in the candidate pool.

10.1.5 Fixed-threshold filter values

In place of the per-worker difference-of-means t-test of [13] (which has effectively no power at $n = 2$ per QC type, see Section 4.2.3), two fixed-threshold filters are applied. Threshold values are fixed in advance and reported here:

- **Bad-reference filter.** Exclude a respondent if *both* of their bad-reference items score $\geq$ the corresponding system output on the targeted rubric dimension. Allowing one miss accommodates accidental misclicks; two misses is signal that the respondent is not discriminating.
- **Repeat-pair filter.** For each repeat pair, compute the absolute score difference per rubric dimension. Exclude a respondent whose mean repeat-pair difference, averaged across the two pairs and the three rubric dimensions, exceeds 25 points (out of 100).

10.1.6 Distribution

The survey is deployed as a Google Form with an Apps Script back-end that implements per-respondent randomised item-subset selection and QC injection. Raw responses dump to a Google Sheet which is exported as CSV for offline aggregation. Distribution is via Mari Carmen Suárez-Figueroa’s network of native-English-speaker contacts within the UPM Easy-to-Read research group’s ecosystem.

The instrument uses 0–100 numeric input fields rather than continuous-slider widgets (Google Forms does not support a true continuous slider). Per Graham et al. [12], the load-bearing property of the protocol is the *continuity* of the rating scale rather than its render-time resolution; integer 0–100 input is faithful to the protocol.

10.2 System Prompts

The prompts below are reproduced verbatim from the deployed system. The idiom set is shown in full; the metaphor prompts differ only in the phenomenon-specific guidance (the metaphor detector follows the MIP procedure, shown below), and are otherwise analogous. All prompts are versioned by SHA-256 hash on each run document for reproducibility.

Idiom detection.

```
You are an expert in linguistics and idiom detection.
Analyze the provided text and detect if it contains any idiomatic expressions, following
    the SemEval-2022 Task 2 guidelines.
```

```
An idiomatic expression (multi-word expression / MWE) is a phrase whose meaning cannot be
    fully derived from the meanings of its individual words. Identify any idiomatic
    expressions in the text.

Return:
- is_figurative: true if any idiomatic expression is present, false otherwise
- figurative_expressions: for each idiom, copy the exact multi-word expression from the
    input verbatim (preserving casing/punctuation).
- explanation: brief reasoning for your detection

For the replacement field, return null as we are only detecting.
```

Idiom explanation (pipeline step 2).

```
You are an expert in linguistics. You are given an English sentence and a list of
    idiomatic expressions that have already been detected in it. Your task is to provide,
    for each detected idiom, a clear literal explanation of what it means in the specific
    context of this sentence.

Each explanation should be a short, plain-English restatement of the idiom's contextual
    meaning - phrased so that it could substitute for the idiom in the sentence while
    preserving the intended meaning. Do not produce a definition in dictionary style;
    produce a substitution-ready paraphrase of the idiom's meaning here.

Return a JSON object with a single field `explanations`, an array with one entry per
    detected idiom:
- expression: the exact idiomatic expression (verbatim from the input list)
- meaning: a clear literal restatement of the idiom's meaning in this specific context
```

Idiom transformation (pipeline step 3).

```
You are an expert in linguistics and Easy-to-Read text adaptation. You will be given an
    English sentence, a list of idiomatic expressions detected in it, and a literal
    meaning for each idiom in this specific context.

Your task: rewrite the FULL sentence shown below the markers, substituting each detected
    idiom with the supplied literal meaning. Keep all non-idiomatic content, sentence
    structure, and word order as close to the original as possible - change only the
    figurative parts.

Output requirements:
- Output ONLY the rewritten sentence - one line of plain English.
- Do NOT include the words "Sentence:", "Output:", "Rewritten:", or any labels.
- Do NOT include explanations, lists, JSON, code blocks, or quotes around the sentence.
```

Monolithic single-prompt baseline (idiom).

```
Rewrite the following sentence in plain, literal English. If it contains any figurative or
    idiomatic expressions, replace them with literal alternatives that preserve the
    original meaning. Otherwise, keep the sentence as is.

Output only the rewritten sentence. Do not provide explanations, lists, or any other
    content.
```

Metaphor detection (MIP procedure).

```
You are an expert in linguistics and metaphor detection.
Analyze the provided text and detect if it contains any metaphorical usage, following the
    VU Amsterdam Metaphor Corpus (VUAMC) guidelines.
```

```
A word is metaphorical if its contextual meaning differs from its basic, most
    concrete/physical meaning. Apply the MIP (Metaphor Identification Procedure):
1. Read the entire text to understand its meaning.
2. For each lexical unit, determine its contextual meaning.
3. Determine if the lexical unit has a more basic meaning (more concrete, body-related, or
    historically older).
4. If the contextual meaning contrasts with the basic meaning but can be understood in
    comparison with it, mark it as metaphorical.

Return:
- is_figurative: true if any word is metaphorical, false otherwise
- figurative_expressions: for each metaphorical word, copy its exact text from the input
    verbatim. List each metaphorical word individually.
- explanation: brief reasoning for your detection
```

10.3 Ethics and Informed Consent

The human-evaluation survey collected no personally identifying information. The only demographic field was a binary native-English-speaker self-attestation (Section 10.1): no name, email, age, or contact detail was recorded, and responses are stored and reported only in aggregate. Participation was voluntary, could be stopped at any time, and individual ratings were optional, so a respondent could submit a partially completed form.

The intro page presented a plain-language task description, an eligibility question, a consent question, and the data-protection statement reproduced below. The statement was supplied by the supervising group and is given verbatim. A respondent who selected “I do not consent” was branched directly to a closing screen and contributed no data.

Eligibility question. *Which of the following describes you?* with the options “I am a native English speaker” and “I am not a native English speaker, but I have lived in an English-speaking country for ten years or more” (a third option ended the form).

Consent question. *I have read the description above and consent to my anonymous responses being used as described*, with the options “I consent” and “I do not consent” (the latter branching to the closing screen).

Data-protection statement (verbatim).

> This questionnaire has been designed to ensure complete anonymity. However, we consider it appropriate to inform you about our privacy policy, in accordance with the provisions of the personal data protection regulations. If you agree to complete the questionnaire, we inform you that you consent to your data being processed by the Universidad Politécnica de Madrid (Ontology Engineering Group), which is responsible for this processing, for the purposes related to the aforementioned research. We also inform you that you may access, rectify and delete your data, as well as exercise other rights, under the terms indicated in the additional information available at the Ontology Engineering Group data-protection page (`oeg.fi.upm.es/protecciondatos.html`).

10.4 Experimental Configuration and Run Index

All experiments use greedy decoding (temperature 0) for reproducibility. The two open-weights models are `meta-llama/Meta-Llama-3.1-8B-Instruct` (FP16, the deliverable) and `casperhansen/llama-3.3-70b-instruct-awq` (the scale comparator), each served with vLLM behind an OpenAI-compatible API. The 70B-AWQ model is served with a maximum context length of 4096 tokens and a GPU memory-utilisation target of 0.90; client-side request concurrency was 16 for the 8B model and 6 for the larger 70B-AWQ model. Every run is persisted under a unique identifier; Table 9 maps each reported result to the run that produced it, so any number in Chapters 5 and 7 can be traced to its source.

Condition / dataset	**8B run**	**70B run**
Detection only (full evaluation set)		
SemEval idiom (3,487)	`8141d880`	`7cbef03f`
VUAMC metaphor (16,202)	`2d322ad0`	`d47a01fb`
Detect-then-replace, v2 (full evaluation set)		
SemEval idiom	`d3a7a15c`	`162b7339`
VUAMC metaphor	`491d332c`	`ffd25507`
Three-step pipeline (full evaluation set)		
SemEval idiom	`b9f31108`	`4bacbfb2`
VUAMC metaphor	`66726606`	`6c4c1fd5`
Replacement metrics, gold-paraphrase subset (200-example slice)		
SemEval idiom, single-call v1	`b2b174ea`	—
SemEval idiom, single-call v2	`d6bc60b0`	`92941662`
SemEval idiom, pipeline	`3551e201`	`56f19b8c`

Table 9: Run index. Detection metrics (sentence and span F1) are computed on the full evaluation sets; the reference-based replacement metrics (BLEU, BERTScore) are computed on the SemEval Task B gold-paraphrase items within the deterministic 200-example slice, as explained in Section 7.

10.5 Per-Respondent Quality Control

Table 10 lists the quality-control outcome for each of the twenty-two returned forms, under the fixed-threshold filters of Section 4.2.3. Two respondents exceed the 25-point repeat-pair threshold and are dropped; no respondent is removed by the bad-reference rule, and twenty are retained for analysis.

Table 10: Per-respondent quality-control outcomes. Eligibility is the self-attestation (native speaker, or non-native resident in an English-speaking country for ten years or more). Repeat-pair |diff| is the mean absolute first-versus-second-showing difference over the two repeat pairs (blank where a respondent did not complete both showings); bad-reference misses count how many of the two bad-reference items the respondent failed to score below the system output. The two respondents exceeding the 25-point repeat-pair threshold are dropped.

Resp.	Eligibility	Items	Rep. |d|	Bad-ref	Verdict
1	Native	20	31.7	1/2 misses	**drop**
2	Resident 10y+	20	4.0	0/2 misses	pass
3	Native	20	11.7	0/2 misses	pass
4	Native	15	–	0/2 misses	pass
5	Resident 10y+	20	9.2	0/2 misses	pass
7	Native	20	1.7	0/2 misses	pass
8	Native	20	6.7	0/2 misses	pass
9	Native	20	9.2	1/2 misses	pass
10	Native	20	25.0	1/2 misses	pass
11	Native	19	10.0	0/2 misses	pass
12	Native	20	25.0	0/2 misses	pass
13	Native	18	30.0	0/1 misses	**drop**
14	Native	20	0.0	0/2 misses	pass
15	Native	20	13.3	0/2 misses	pass
20	Resident 10y+	20	22.5	0/2 misses	pass
22	Native	11	–	0/1 misses	pass
23	Native	20	3.3	0/2 misses	pass
24	Native	19	8.3	0/2 misses	pass
25	Native	17	–	0/2 misses	pass
26	Native	15	–	1/2 misses	pass
27	Native	20	13.3	0/2 misses	pass
28	Native	19	5.8	0/2 misses	pass

10.6 Per-Item Human-Evaluation Scores

Table 11 gives the raw per-item means behind the aggregate results of Section 7.6, for all twenty-seven sources under each of the three systems. These are the disaggregated form of Table 6.

Table 11: Per-item human-evaluation raw means (0–100) for all 27 sources under the three systems. Each cell is the mean across the respondents who rated that (source, system) pair; the composite is the mean of the three per-annotator-standardised dimensions. Source identifiers match the survey pool.

Source	**System**	**Gram.**	**Mean.**	**Simp.**	**Comp.** z
src_00	8B agentic	87.5	82.5	46.2	+0.161
	8B monolithic	95.0	96.2	72.5	+0.475
	70B agentic	88.8	86.2	55.0	+0.069
src_01	8B agentic	91.2	87.5	95.0	+0.576
	8B monolithic	82.0	77.0	63.0	+0.282
	70B agentic	83.8	85.0	53.8	+0.002
src_02	8B agentic	86.7	88.3	90.0	+0.569
	8B monolithic	49.0	40.0	34.0	-1.145
	70B agentic	52.5	33.8	23.8	-0.936
src_03	8B agentic	85.0	69.0	52.0	-0.364
	8B monolithic	91.2	37.5	37.5	-0.476
	70B agentic	57.5	83.8	32.5	-0.332
src_04	8B agentic	89.6	86.0	73.0	+0.256
	8B monolithic	80.0	64.0	39.0	-0.340
	70B agentic	33.3	33.3	16.7	-1.380
src_05	8B agentic	83.3	100.0	53.3	+0.315
	8B monolithic	65.0	60.0	65.0	-0.117
	70B agentic	89.0	79.0	49.0	+0.194
src_06	8B agentic	86.2	62.5	60.0	+0.153
	8B monolithic	88.8	82.5	36.2	-0.007
	70B agentic	83.3	80.0	43.3	+0.059
src_07	8B agentic	83.3	80.0	63.3	+0.260
	8B monolithic	61.7	30.0	46.7	-0.741
	70B agentic	75.0	27.5	40.0	-0.827
src_08	8B agentic	76.0	52.0	46.0	-0.312
	8B monolithic	87.5	45.0	20.0	-0.720
	70B agentic	88.3	21.7	25.0	-0.904
src_09	8B agentic	76.2	78.8	45.0	-0.042
	8B monolithic	92.0	65.0	55.0	+0.177
	70B agentic	87.5	82.5	47.5	+0.062
src_10	8B agentic	80.0	70.0	70.0	-0.132
	8B monolithic	91.2	88.8	85.0	+0.564
	70B agentic	69.0	63.2	48.0	-0.312

Source	System	Gram.	Mean.	Simp.	Comp. z
src_11	8B agentic	88.8	85.0	76.2	+0.297
	8B monolithic	82.5	77.5	72.5	+0.293
	70B agentic	90.0	75.0	67.5	+0.314
src_12	8B agentic	78.3	63.3	50.0	-0.293
	8B monolithic	88.8	51.2	72.5	+0.011
	70B agentic	93.0	89.0	64.0	+0.000
src_13	8B agentic	98.3	75.0	50.0	+0.238
	8B monolithic	98.0	82.0	78.0	+0.630
	70B agentic	92.5	68.8	58.8	-0.033
src_14	8B agentic	51.0	60.0	24.0	-0.928
	8B monolithic	93.8	96.2	91.2	+0.561
	70B agentic	97.5	97.5	70.0	+0.613
src_15	8B agentic	89.0	93.0	34.0	+0.163
	8B monolithic	90.0	95.0	80.0	+0.658
	70B agentic	93.5	93.5	91.0	+0.823
src_16	8B agentic	90.0	100.0	87.5	+0.609
	8B monolithic	95.0	95.0	85.0	+0.660
	70B agentic	75.0	70.0	36.2	-0.372
src_19	8B agentic	90.0	93.3	85.3	+0.507
	8B monolithic	55.0	55.0	70.0	+0.034
	70B agentic	92.6	87.8	52.0	+0.213
src_20	8B agentic	70.0	100.0	70.0	+0.312
	8B monolithic	91.2	87.5	57.5	+0.039
	70B agentic	90.0	93.3	26.7	+0.104
src_21	8B agentic	90.0	88.2	63.0	+0.127
	8B monolithic	90.0	84.0	59.0	+0.410
	70B agentic	97.5	97.5	87.5	+0.488
src_23	8B agentic	100.0	100.0	100.0	+0.725
	8B monolithic	85.0	61.2	80.0	+0.217
	70B agentic	100.0	100.0	66.7	+0.513
src_24	8B agentic	58.8	88.8	57.5	-0.120
	8B monolithic	95.0	93.8	83.8	+0.627
	70B agentic	82.0	97.0	58.0	+0.067
src_25	8B agentic	70.0	100.0	53.3	-0.032
	8B monolithic	88.0	84.0	60.0	+0.260
	70B agentic	87.5	75.0	75.0	+0.463
src_26	8B agentic	95.0	97.5	60.0	+0.524

Source	System	Gram.	Mean.	Simp.	Comp. z
	8B monolithic	76.2	32.5	67.5	-0.315
	70B agentic	66.7	81.7	45.0	-0.299
src_27	8B agentic	98.0	97.0	47.0	+0.361
	8B monolithic	93.0	86.0	86.0	+0.527
	70B agentic	95.0	82.2	72.5	+0.407
src_28	8B agentic	97.5	65.0	60.0	+0.152
	8B monolithic	87.5	72.5	43.8	-0.200
	70B agentic	41.0	89.0	58.8	-0.296
src_29	8B agentic	60.0	53.3	83.3	+0.158
	8B monolithic	95.0	95.0	86.7	+0.691
	70B agentic	37.5	86.2	51.7	-0.555

10.7 Sample of Collected Alternative Replacements

Respondents could add a free-text alternative to any item, and these alternatives form part of the dataset released with this thesis (Section 7.7). Table 12 shows a representative sample. The recurring pattern is compression: where the system substitutes a literal gloss, the human reader reaches for a shorter, plainer phrase or removes part of the gloss.

Detected expression	System	Respondent's proposed alternative
smoking gun	8B agentic	"strong evidence against the accused"
brush the incident aside; chalk it up to ancient history	70B agentic	"and worry about it later"
black operation	8B agentic	"or some shady mission"
all-rounders	8B monolithic	"business experts"
speed trap	70B agentic	would drop "or activity" from the gloss

Table 12: A sample of the free-text alternative replacements supplied by respondents, illustrating the compression pattern discussed in Section 7.7.